ERODE SENGUNTHAR

ENGINEERING COLLEGE

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Lab Code/Name :……

Semester :……

Faculty Incharge Head of the Department

Submitted for the End Semester Practical Examination held On……

Internal Examiner *External Examiner*

INDEX

Ex. No	Date	LIST OF EXPERIMENTS	Page No.	Marks	Sign
1		Implementation to gather information from any PC's connected to the LAN using who is, port scanners, network scanning, Angry IP scanners etc.			
2		Implementation of Steganography			
3		Implementation of Mobile Audit and generate the report of the existing Artifacts.			
4		Implementation of IT Audit, malware analysis and Vulnerability assessment and generate the report.			
5		Implementation of Cyber Forensics tools for Disk Imaging, Data acquisition, Data extraction and Data Analysis and recovery.			
6		Perform mobile analysis in the form of retrieving call logs, SMS log, all contacts list using the forensics tool like SAFT			
7		Implementation to identify web vulnerabilities, using OWASP project.			
8		Configuration and Experimentation of Firewall and Proxy Services using iptables and Squid on Kali Linux			

Ex. No. 1	Implementation to gather information from any PC's connected to the LAN using who is, port scanners, network scanning, Angry IP scanners etc.

AIM:

To gather information about PCs connected to the LAN using tools like who is, port scanners, network scanning, and Angry IP Scanner.

OBJECTIVE:

To understand network topology and retrieve detailed information about connected devices, including IP addresses, open ports, and services.

REQUIREMENTS:

- Kali Linux installed on VirtualBox
- Tools: whois, nmap, Angry IP Scanner
- A PC connected to the same LAN
- Internet connection

THEORY:

Who is

Who is searches for an object in a WHOIS database. WHOIS is a query and response protocol widely used for querying databases that store the registered users of an Internet resource, such as a domain name or an IP address block, but it is also used for a wider range of other information. Most modern versions of whois try to guess the right server to ask for the specified object. If no guess can be made, whois will connect to whois.networksolutions.com for NIC handles or whois.arin.net for IPv4 addresses and network names.

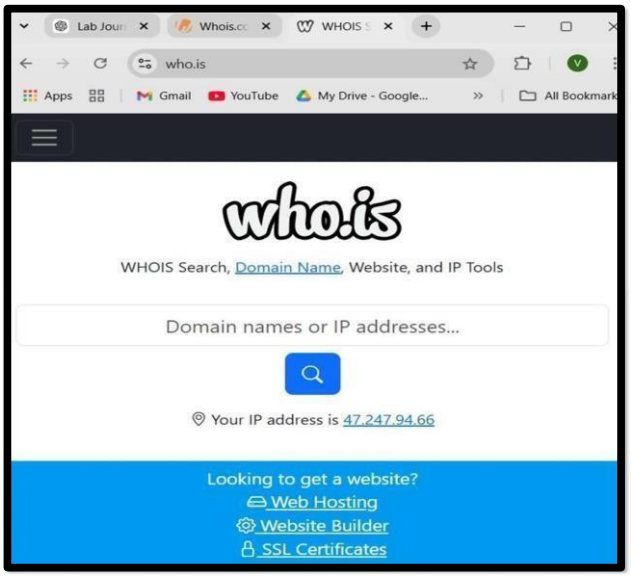
Examples:

- Obtaining the domain WHOIS record for computersolutions.com
- WHOIS record by IP querying
- Querying WHOIS in Google search engine

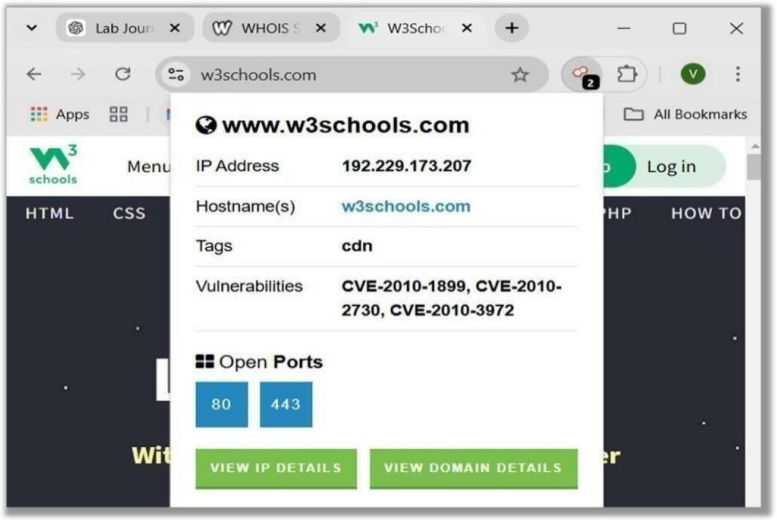
To use the Whois lookup tool, just enter the domain name whose information you'd like to view into the search field on the Whois main page. You can retrieve key data about a domain in this way, including availability, domain owner lookup, and creation and expiration details. If you own multiple domains, it can be helpful to download exportable lists from the tool to analyze large amounts of domain data.

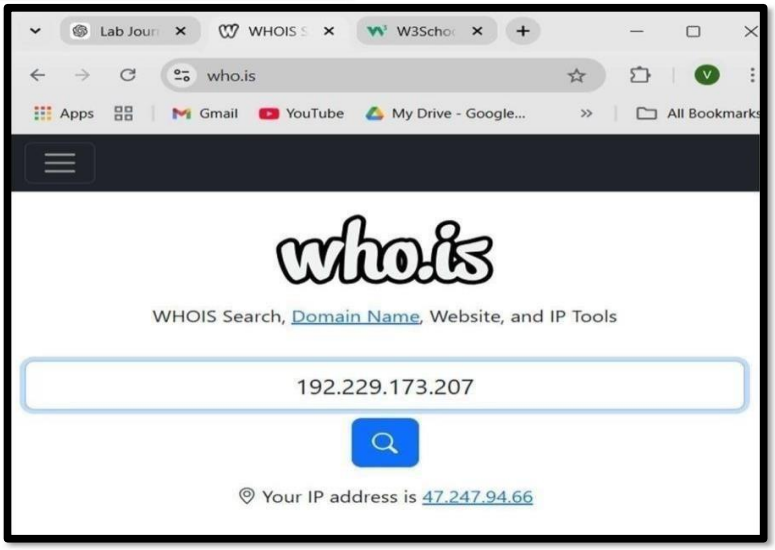
Implementation:

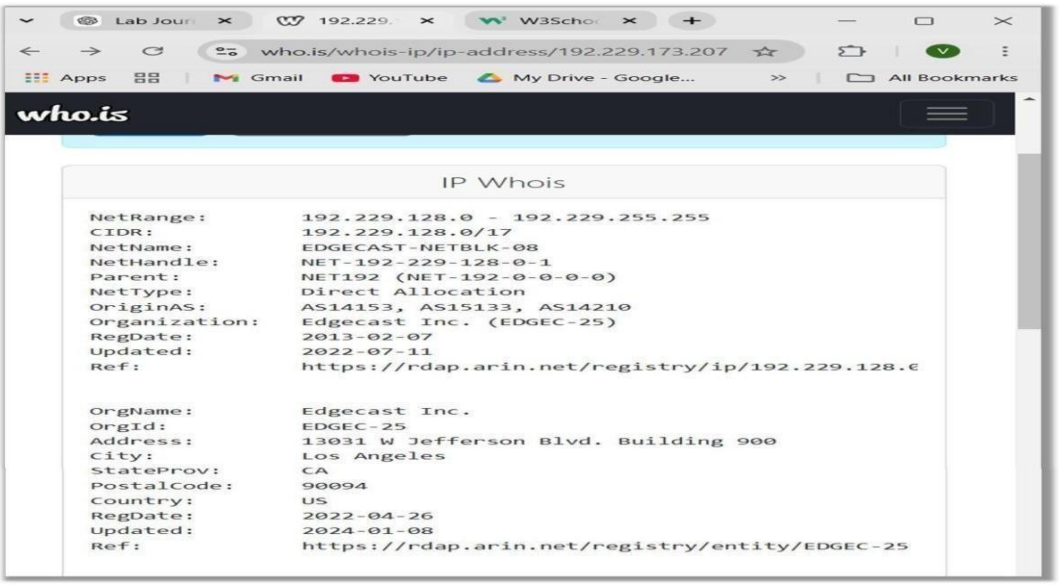
Step 1: Search whois in the web browser Step 2: Open the whois tool



Step 3: Enter the domain name or IP address of the device whose information you want to gather using a plugin called Shodan.







Port Scanners

- Port scanners are essential tools in network security and administration.
- They identify open ports on a network and determine the services running on them.

- Each port corresponds to a specific service (e.g., HTTP runs on port 80).
- Detecting open ports helps administrators secure the network and identify vulnerabilities.
- Zenmap is the official graphical user interface (GUI) for Nmap, a popular open-source network scanner.
- Zenmap provides a user-friendly way to perform network scans, visualize results, and save configurations for reuse.

Features of Zenmap:

- Performs a variety of scans like Quick Scan, Intense Scan, and Ping Scan.
- Visualizes the network topology graphically.
- Offers customizable profiles for scan automation.
- Displays results in multiple formats, including plain text, XML, and graphs.

Implementation Steps (Using Zenmap):

Launch Zenmap: Open Zenmap on your computer.

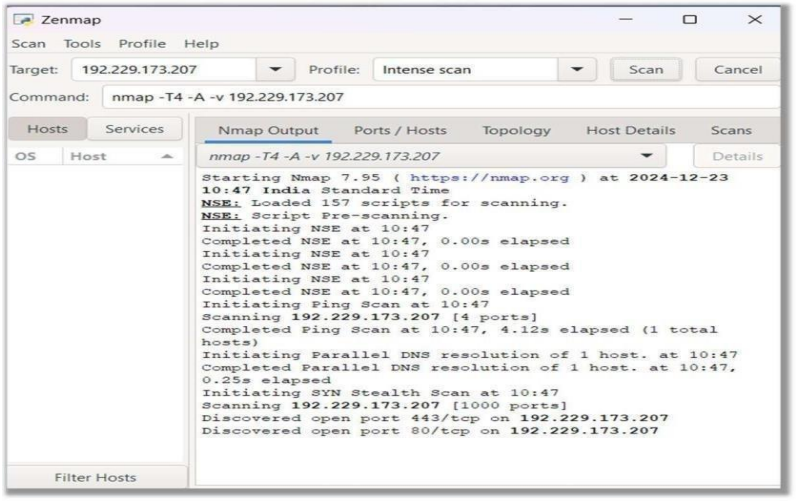
Target Selection: In the "Target" field, enter the IP address or IP range to scan (e.g., 192.168.1.0/24).

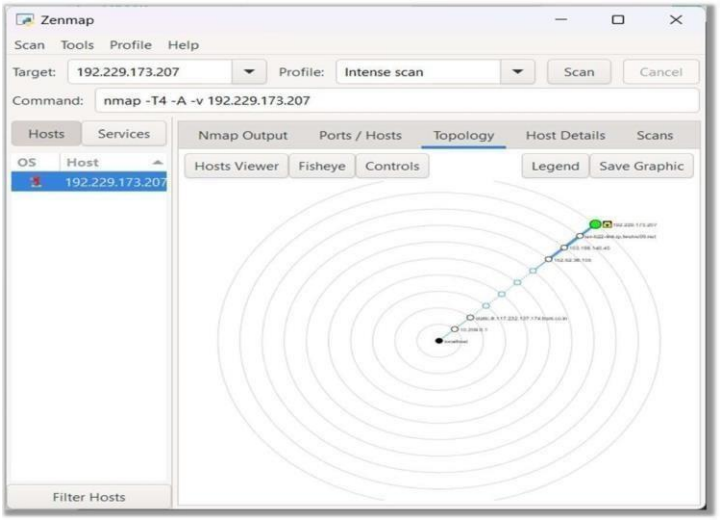
Profile Selection: Choose a predefined scan profile from the "Profile" dropdown menu (e.g., Quick Scan or Intense Scan). For detailed results, select "Intense Scan."

Start Scan: Click the "Scan" button to initiate the process. Zenmap will begin scanning the target and display progress.

Review Results: Switch to the "Nmap Output" tab to view detailed information about open ports, services, and OS details. Check the "Topology" tab for a graphical representation of the network.

Save Results: Save the scan results for future reference by clicking "Scan > Save Scan."





Angry IP Scanner

Angry IP Scanner is a lightweight and user-friendly network scanner. It scans IP addresses and ports to detect active hosts and retrieve additional information like MAC addresses and hostnames. Unlike Zenmap, Angry IP Scanner is designed for simplicity and quick scans, making it suitable for smaller networks or fast reconnaissance.

Features of Angry IP Scanner:

- Scans a range of IP addresses and detects active hosts.
- Provides additional details like MAC addresses and NetBIOS information.
- Supports exporting results in multiple formats (CSV, TXT, etc.).
- Allows adding custom "fetchers" to extend its functionality.

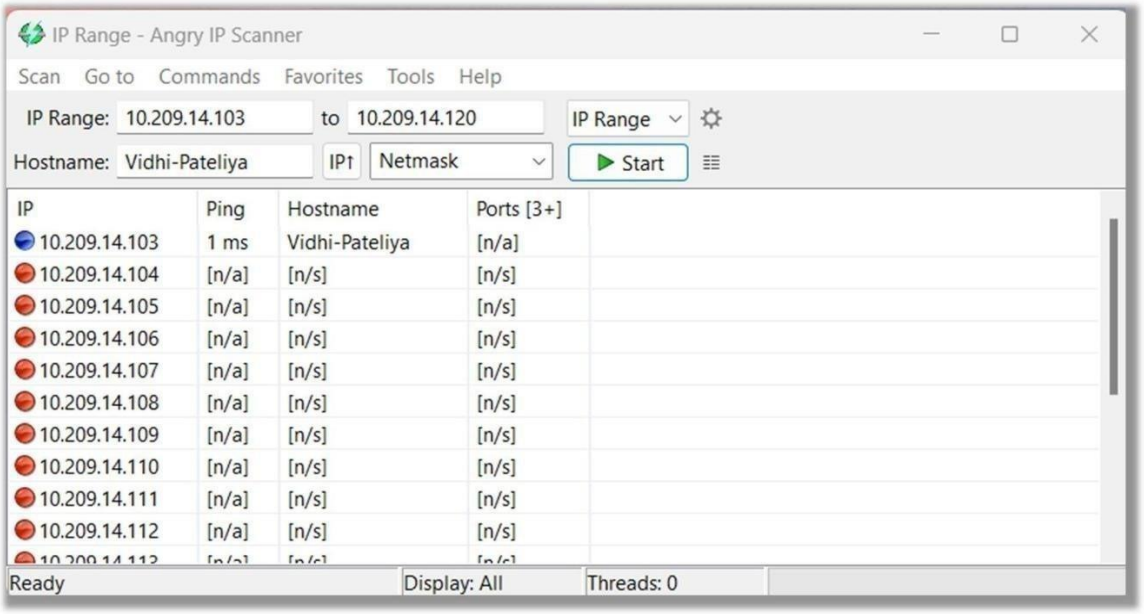
Implementation Steps (Using Angry IP Scanner):

Launch Angry IP Scanner: Open the application.

Set IP Range: In the "IP Range" fields, enter the starting and ending IP addresses to scan (e.g., 192.168.1.1 to 192.168.1.254).

Add Fetchers: Navigate to "Tools > Fetchers" to add extra details like MAC addresses, NetBIOS names, or open ports.

Start Scanning: Click the "Start" button to initiate the scan. Angry IP Scanner will ping each IP and check for active hosts.



Review Results: Active hosts are marked with green dots, while inactive ones are marked red. Additional information like open ports, MAC addresses, and hostnames is displayed in the results table.

Save or Export Results: Save the scan results by clicking "File > Save As." Export data in formats like CSV or TXT for documentation.

RESULT:

Thus the information gathering process on the LAN was successfully performed using WHOIS, Nmap (Zenmap), and Angry IP Scanner. Details such as IP addresses, open ports, active hosts, and running services were identified and verified successfully.

Ex. No. 2	Implementation of Steganography

AIM:

To implement Image Steganography using Open Stego software and an online steganography tool.

THEORY:

What is Steganography?

Steganography is the technique of hiding secret data within an ordinary, non-secret, file or message in order to avoid detection; the secret data is then extracted at its destination. The use of steganography can be combined with encryption as an extra step for hiding or protecting data. The word steganography is derived from the Greek words steganos (meaning hidden or covered) and the Greek root graph (meaning to write).

Steganography can be used to conceal almost any type of digital content, including text, image, video or audio content; the data to be hidden can be hidden inside almost any other type of digital content. The content to be concealed through steganography -- called hidden text -- is often encrypted before being incorporated into the innocuous-seeming cover text file or data stream. If not encrypted, the hidden text is commonly processed in some way in order to increase the difficulty of detecting the secret content.

Examples of Steganography:

Steganography is practiced by those wishing to convey a secret message or code. While there are many legitimate uses for steganography, malware developers have also been found to use steganography to obscure the transmission of malicious code.

Forms of steganography have been used for centuries and include almost any technique for hiding a secret message in an otherwise harmless container. For example, using invisible ink to hide secret messages in otherwise inoffensive messages; hiding documents recorded on microdot -- which can be as small as 1 millimeter in diameter -- on or inside legitimate-seeming correspondence; and even by using multiplayer gaming environments to share information.

Advantages of steganography over cryptography:

Steganography is distinct from cryptography, but using both together can help improve the security of the protected information and prevent detection of the secret communication. If steganographically-hidden data is also encrypted, the data may still be safe from detection -- though the channel will no longer be safe from detection. There are advantages to using steganography combined with encryption over encryption-only communication.

The primary advantage of using steganography to hide data over encryption is that it helps obscure the fact that there is sensitive data hidden in the file or other content carrying the hidden text. Whereas an encrypted file, message or network packet payload is clearly marked and identifiable as such, using steganographic techniques helps to obscure the presence of the secure channel.

PROCEDURE:

Image Steganography using OpenStego - an open-source steganography software:

Implementing image steganography using OpenStego involves following steps:

Install OpenStego: First, you need to download and install OpenStego on your system. You can find the installation instructions on the OpenStego website or GitHub repository.

Prepare your files: You'll need two files: the cover image (the image you want to hide the data in) and the secret message or file you want to hide. Make sure the message file is in a supported format (text, image, etc.)

Launch OpenStego: Open the OpenStego application.

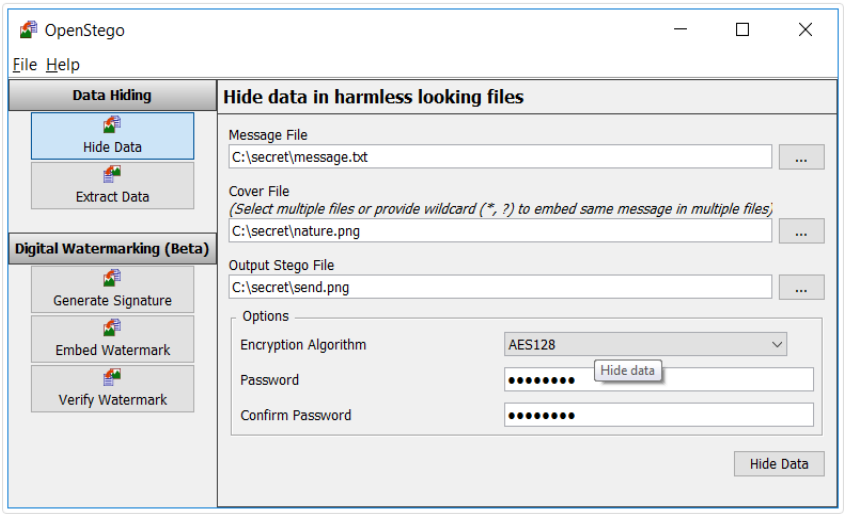
1. For hiding data, select Hide Data option:

4a) Select Message file i.e. (the secret message or file).

4b) Select Cover file i.e. the image in which data is to be hidden. 4c) Select output stego file where output is to be saved.

4d) Click on Hide Data option.

Hide data



5)For Extracting Data:

5a) Select input stego file.

5b) Select output folder for message file. 5c) Click on Extract data.

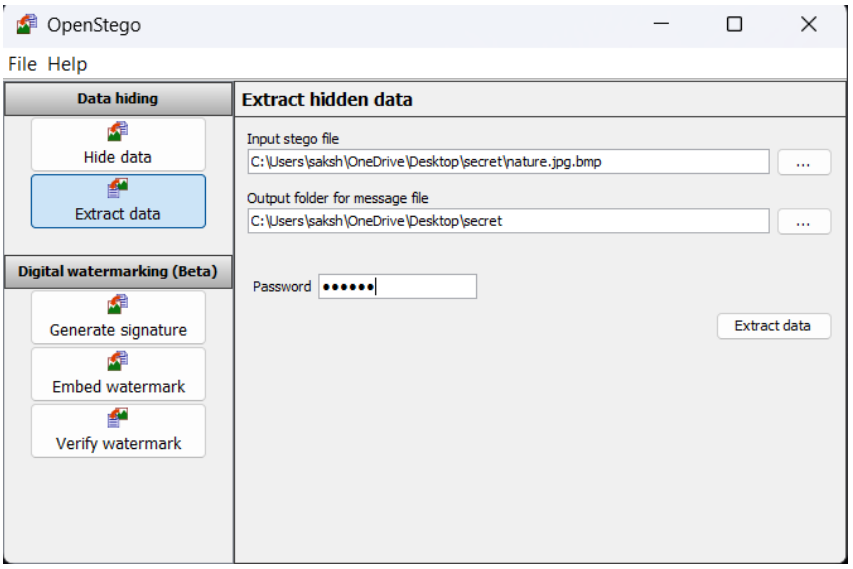


Image Steganography using an online tool:

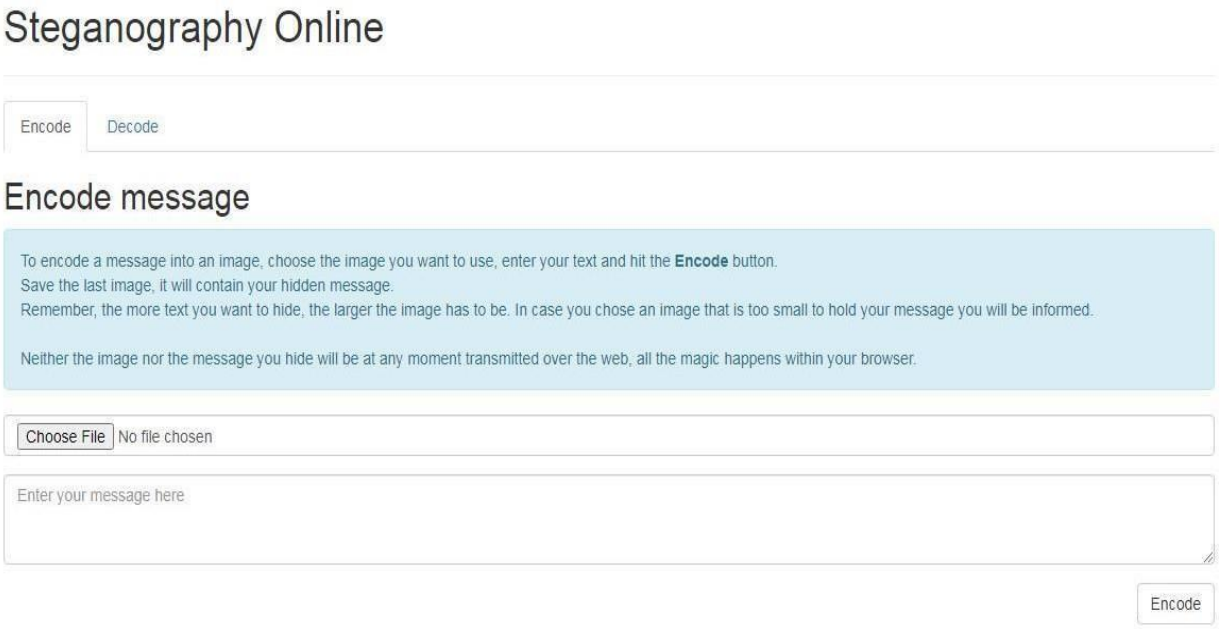
1. Open an online image steganography tool using the link: https://stylesuxx.github.io/steganography/

2. To encode the data:

2a) Select Encode tab.

2b) Click on Choose File to select the cover image. 2c) Type the secret message in the space given.

2d) Click on Encode.

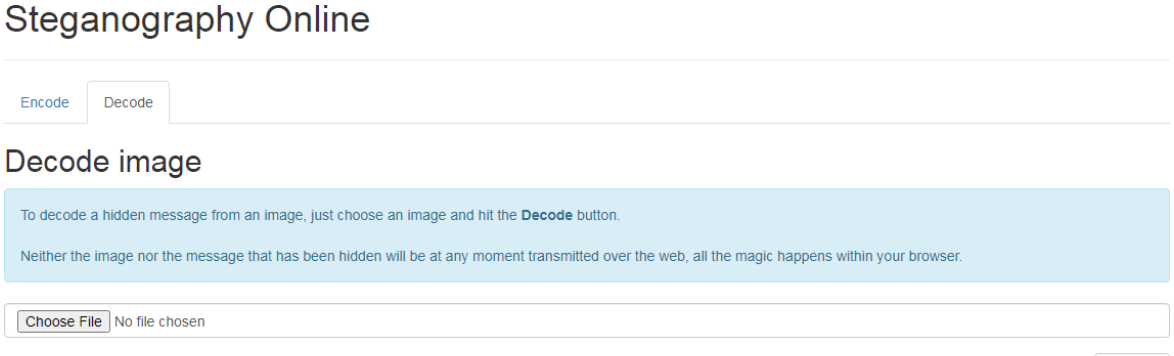


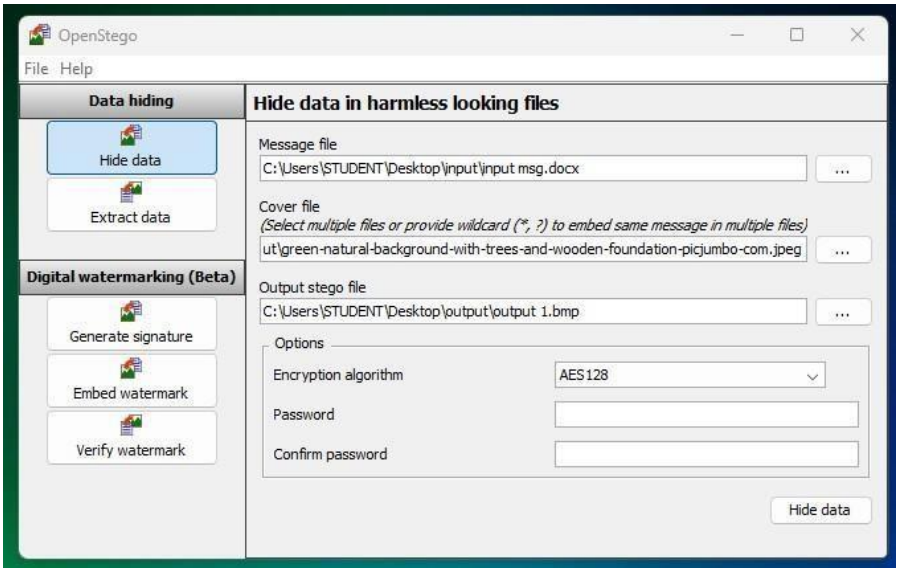
To Decode the data:

a) Select Decode tab.

b) Click on Choose File and select the encrypted file.

c) Click on Decode.





Steganography Online

Encode | Decode

Encode message

To encode a message into an image, choose the image you want to use, enter your text and hit the **Encode** button.
Save the last image, it will contain your hidden message.
Remember, the more text you want to hide, the larger the image has to be. In case you chose an image that is too small to hold your message you will be informed.

Neither the image nor the message you hide will be at any moment transmitted over the web, all the magic happens within your browser.

Choose file leaves-4235968_1280.jpg

Dedicated to the passionate learner of network security.

Encode

Binary representation of your message

0100010001100101011001000110100101100011011000010111010001100101011001000010000001110100011011110010000001110100011010000110010100100000011100
0001100001011100110111001101101001011011110111011100110000101110100011001010011000000011011000110010101100001011100100110111001100101011100100010

Original



Normalized



Steganography Online

Encode | Decode

Decode image

To decode a hidden message from an image, just choose an image and hit the Decode button.

Neither the image nor the message that has been hidden will be at any moment transmitted over the web, all the magic happens within your browser.

Choose file stegano1.png

Decode

Hidden message

Dedicated to the passionate learner of network security.

Input



RESULT:

Thus the Image steganography was successfully implemented using OpenStego and an online steganography tool. The secret data was embedded into the cover image and later extracted without any data loss. This demonstrated secure data hiding techniques and validated the effectiveness of steganography were verified successfully.

Ex. No. 3	Implementation of Mobile Audit and generate the report of the existing Artifacts.

AIM:

To Implement Mobile Audit and generate the report of the existing Artifacts.

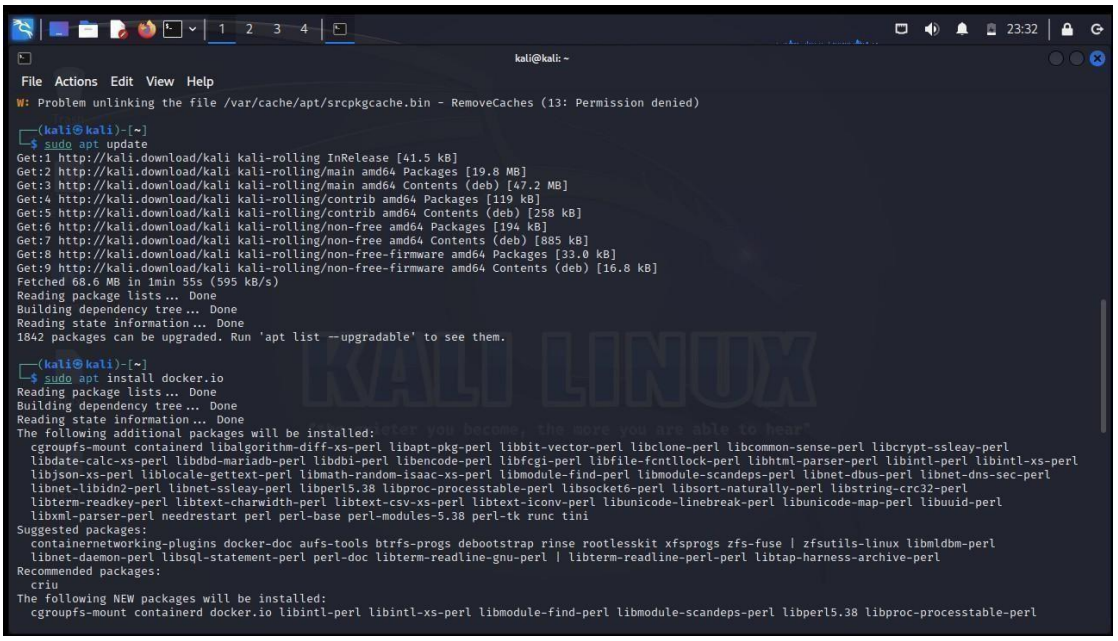
PROCEDURE:

1. Update APT:

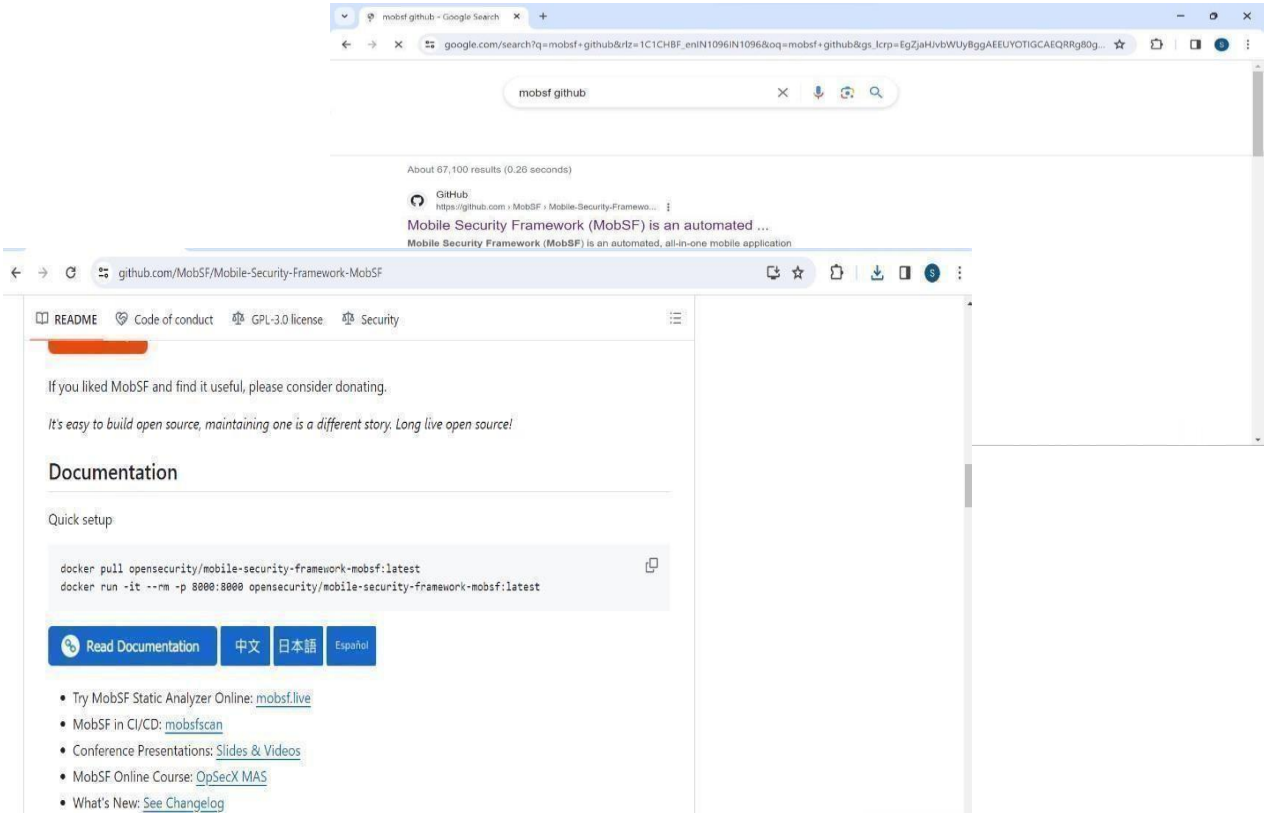
sudo apt update

2. Install docker:

sudo apt install docker.io

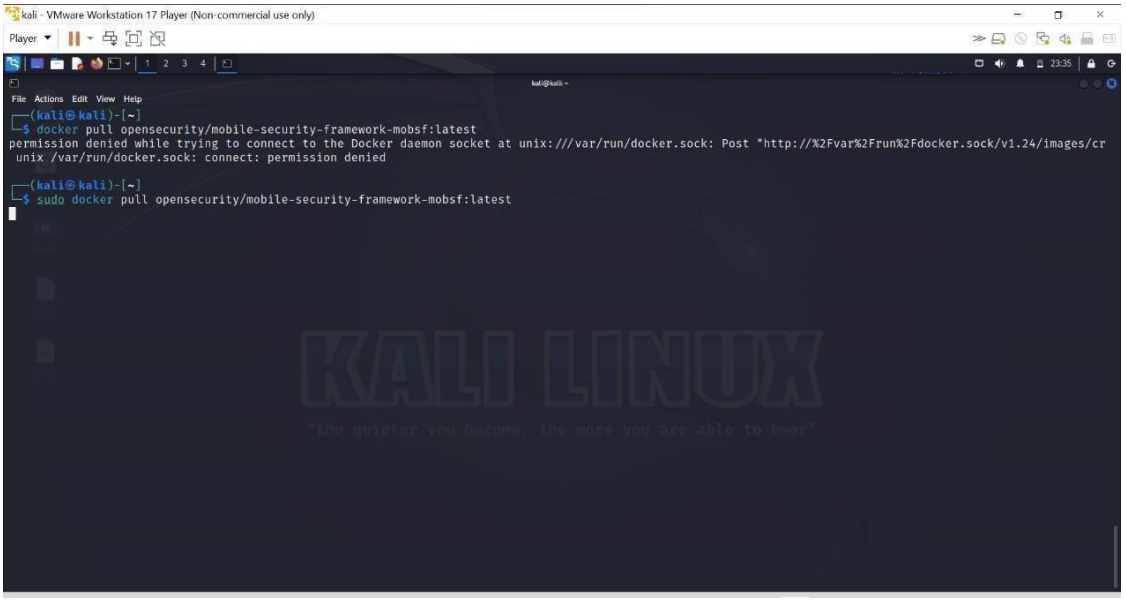


3. Search mobsf github on your web browser.

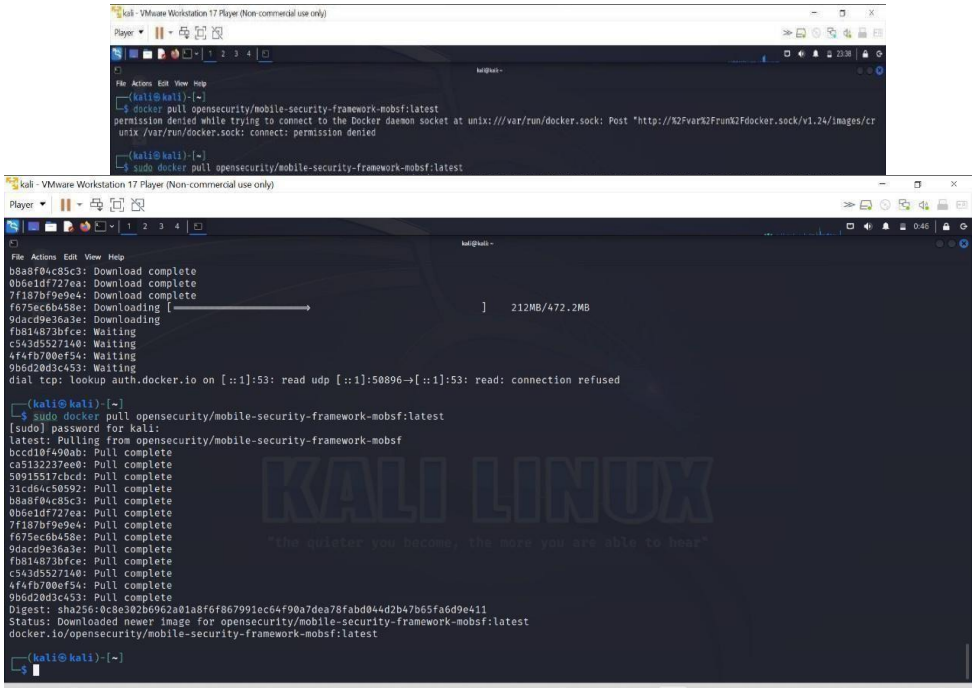


Next, download MobSF Docker image from https://hub.docker.com/r/opensecurity/mobile-security-framework-mobsf/ with the following command:

docker pull opensecurity/mobile-security-framework-mobsf:latest

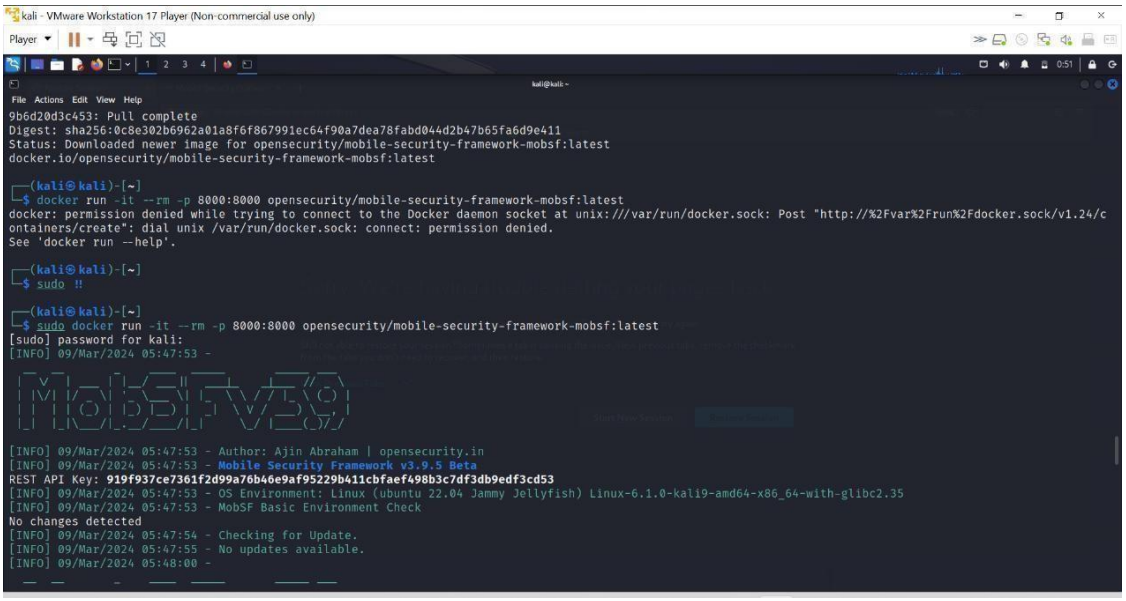


Once you issue the command, you would notice the following output on your console. This signifies that the docker image for MobSF is being downloaded. Once completed, the following message will appear.



Now that the docker image is downloaded, the image can be run with the following command: docker run -it -p 8000:8000 opensecurity/mobile-security-framework-mobsf

Once done, you would see message "Listening at: http://0.0.0.0:8000". This signifies the URL from which one can access MobSF.



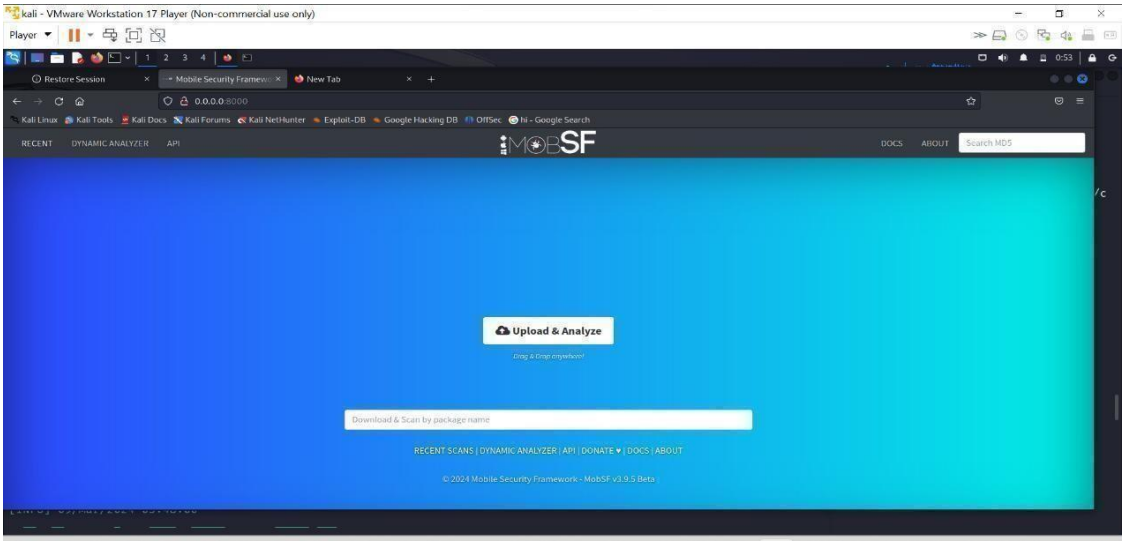
Our MobSF framework is ready for us to conduct static analysis for APK files. For our testing, we take Facebook Lite's APK file.



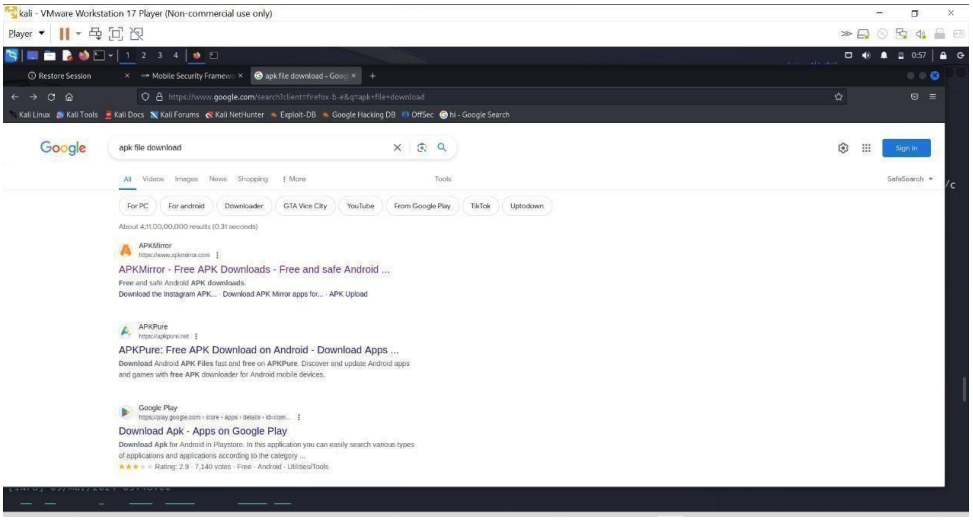
Once you upload the APK file, you would notice the following output being printed on our terminal.

Once the analysis is complete, the browser will show assessment details, such as application description, Android permission analysis, code analysis, CVSS grading, malware analysis etc.

You also notice a brief overview of the application you have reviewed. You can also the report for same. Our MobSF framework is ready for us to conduct static analysis for APK files.



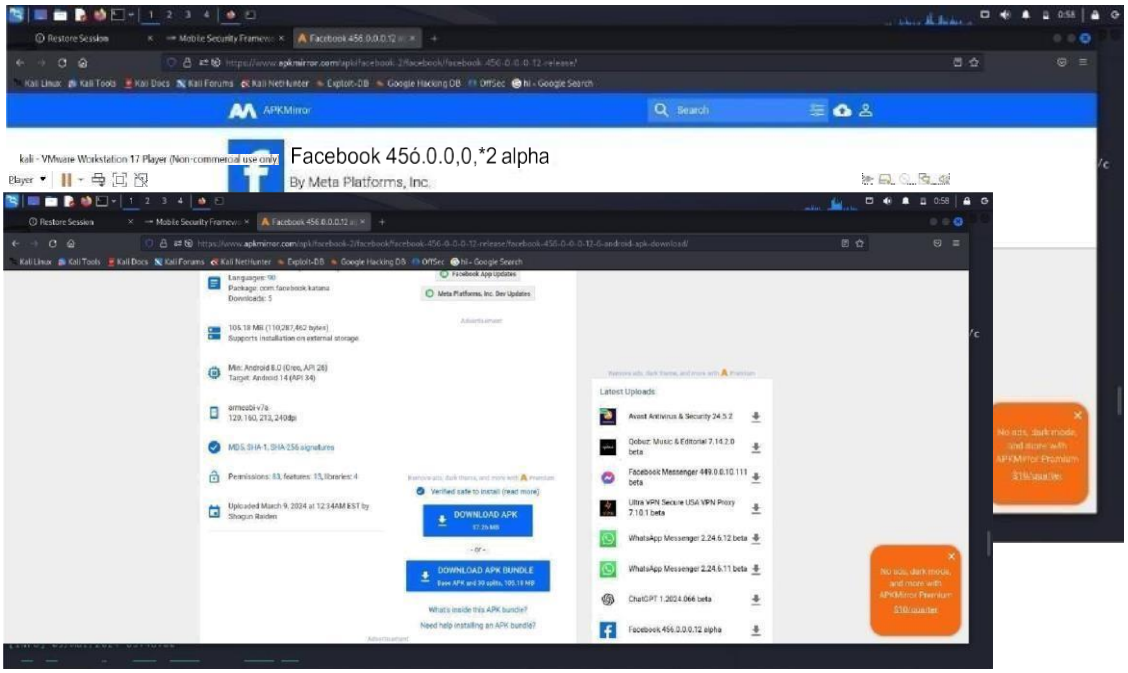
For our testing, we take Facebook Lite's APK file.



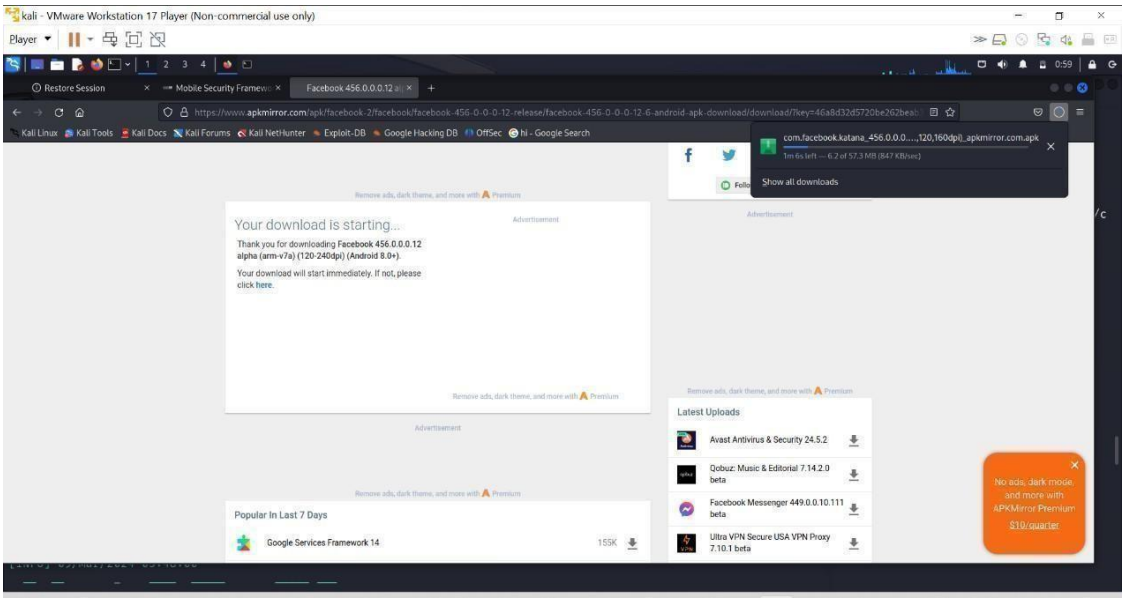
Cyber Security Laboratory

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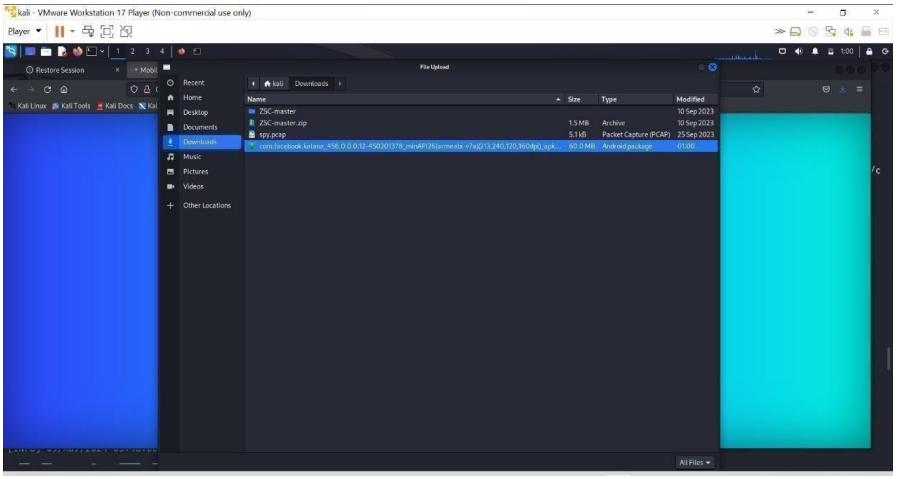
1

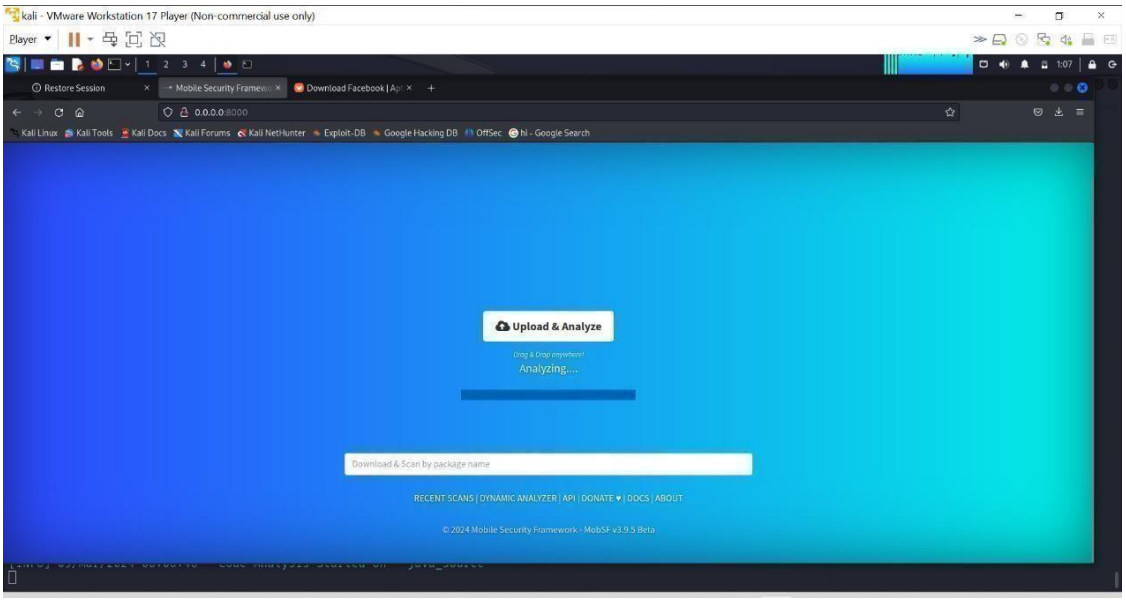


n

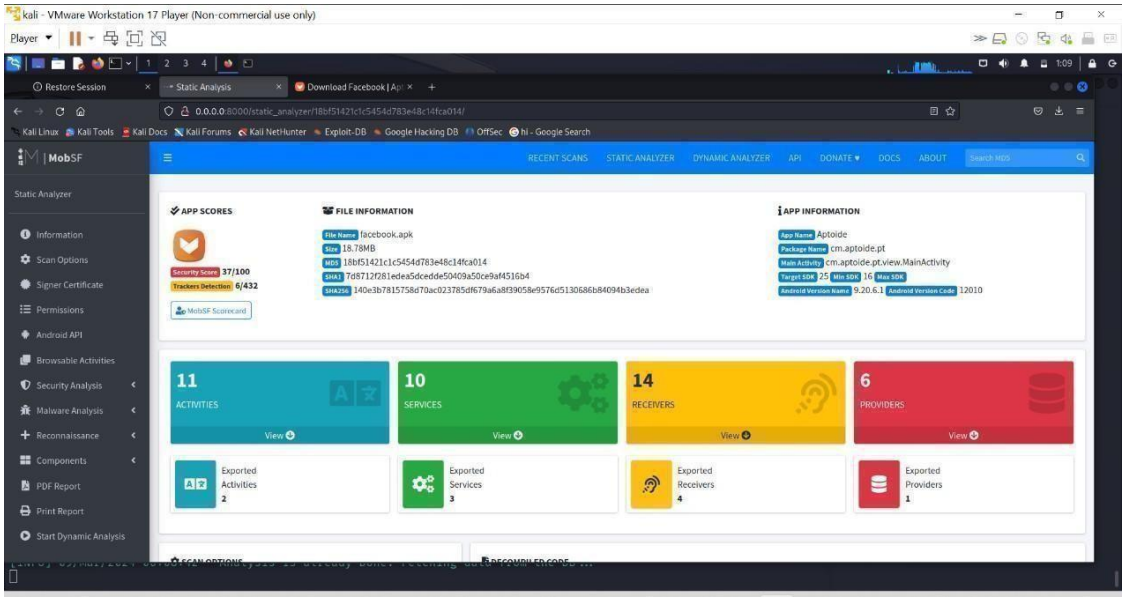


Once you upload the APK file, you would notice the following output being printed on our terminal:

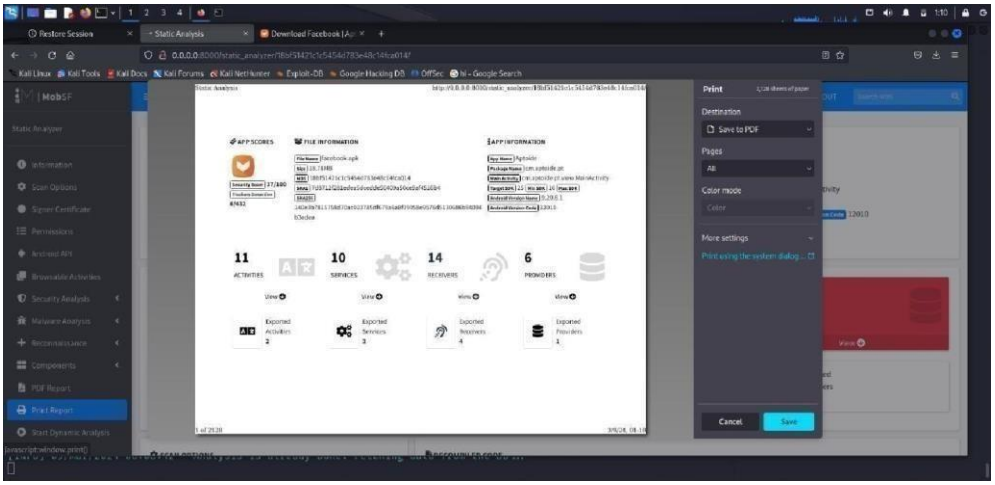




Once the analysis is complete, the browser will show assessment details, such as application description, Android permission analysis, code analysis, CVSS grading, malware analysis etc.



You also notice a brief overview of the application you have reviewed. You can also the report for same.



RESULT:

Thus the Mobile security analysis was successfully conducted using the MobSF framework. The APK file was analyzed, and detailed reports including permissions, vulnerabilities, code analysis, and risk assessment were generated. The tool effectively identified potential security issues in the mobile application was verified successfully.

Ex. No. 4	Implementation of IT Audit, malware analysis and Vulnerability assessment and generate the report.

AIM:

To Implement IT Audit, malware analysis and Vulnerability assessment and generate the report.

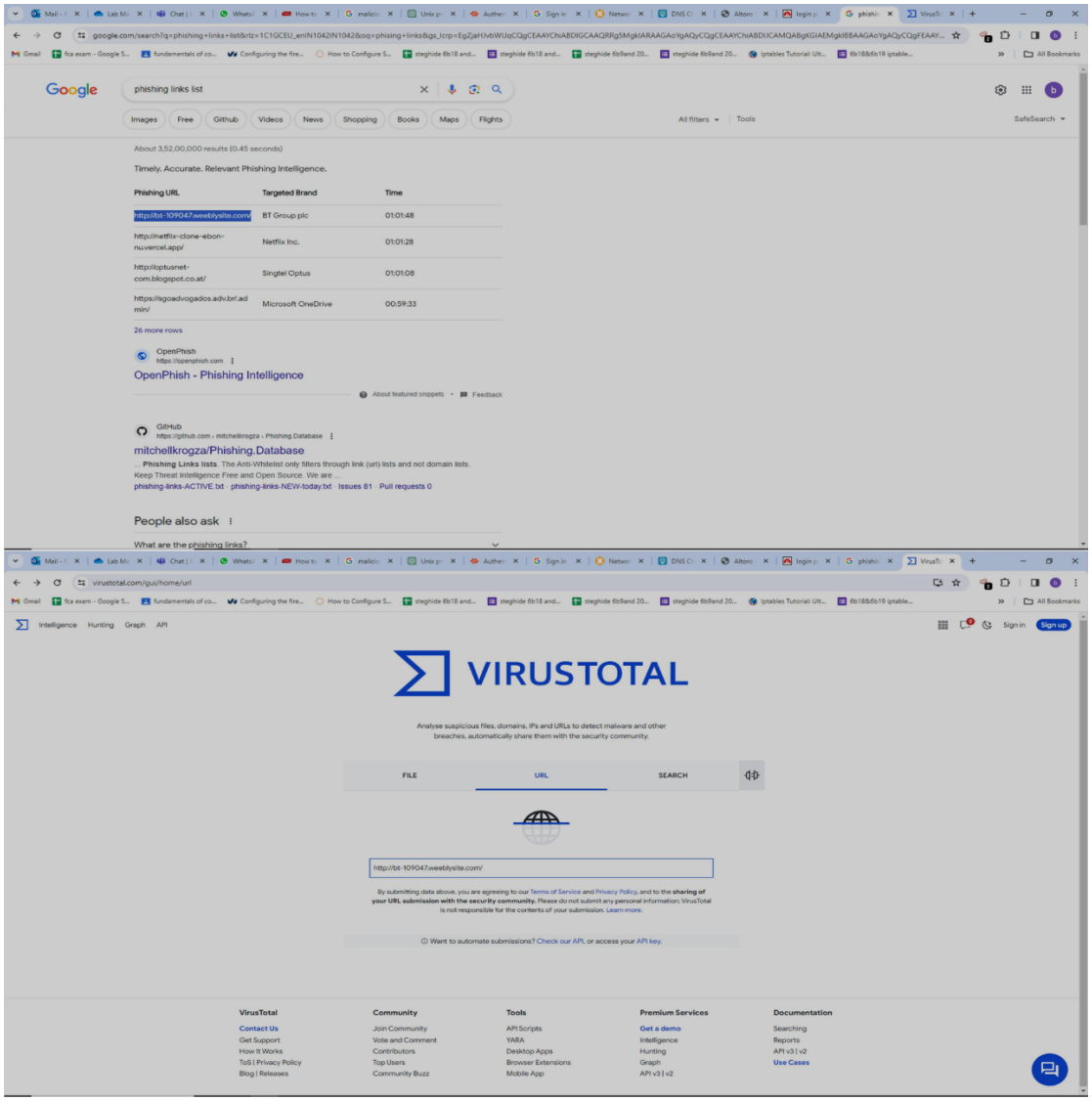
OBJECTIVE:

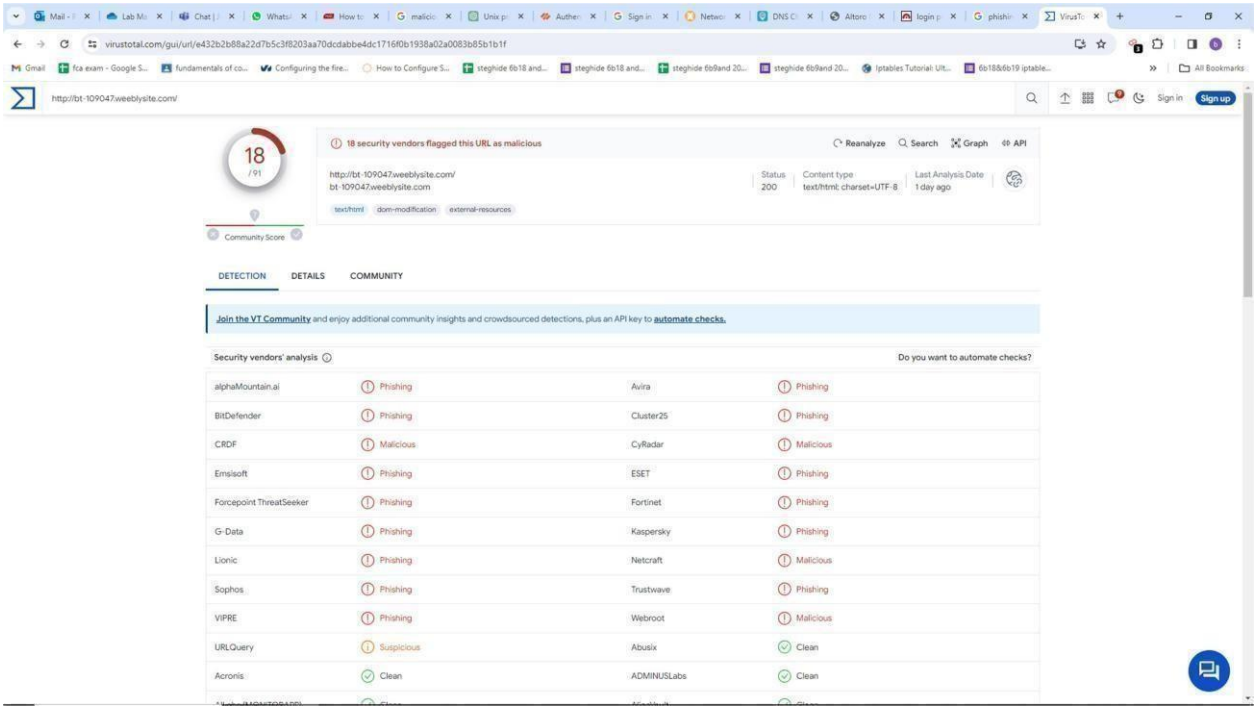
To know how to find vulnerabilities by using NESSUS.

REQUIREMENTS:

Laptop, Kali linux, Nessusd package in kali

MALWARE ANALYSIS





NESSUS:

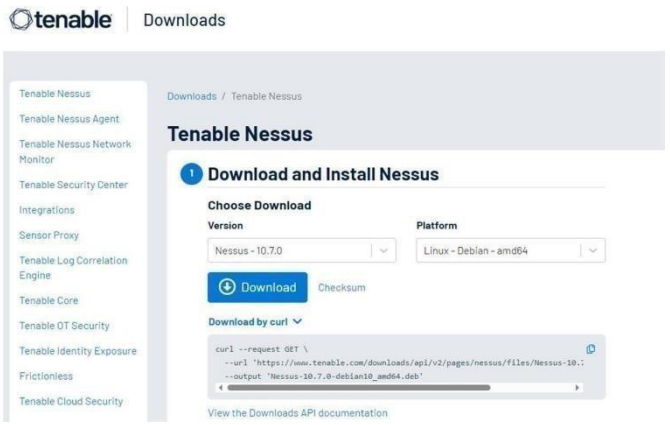
Nessus, developed by Tenable Inc, is a widely-used open-source vulnerability scanner. It offers a paid subscription, Nessus Professional, as well as a free version, Nessus Essentials, which is limited to 16 IP addresses per scanner.

Nessus provides a range of services, including vulnerability assessments, network scans, web scans, asset discovery, and more, to aid security professionals, penetration testers, and other cybersecurity enthusiasts in proactively identifying and mitigating vulnerabilities in their networks.

How to install Nessus in Kali

Unlike many security tools, Nessus doesn't come installed on Kali Linux. But it is very easy to download and install. Follow these steps to install Nessus on your Kali:

1. Download the Nessus package for Debian on the Nessus website and make sure you set the Platform to Linux- Debian-amd64.



2. When it's finished downloading, open your Linux terminal and navigate to the location you downloaded the Nessus file to.
3. Install Nessus using this command:

```
┌──(root㉿kali)-[~]
└─# systemctl status nessusd.service
Unit nessusd.service could not be found.

┌──(root㉿kali)-[~]
└─# systemctl status nessusd
Unit nessusd.service could not be found.

┌──(root㉿kali)-[~]
└─# nessusd
nessusd: command not found

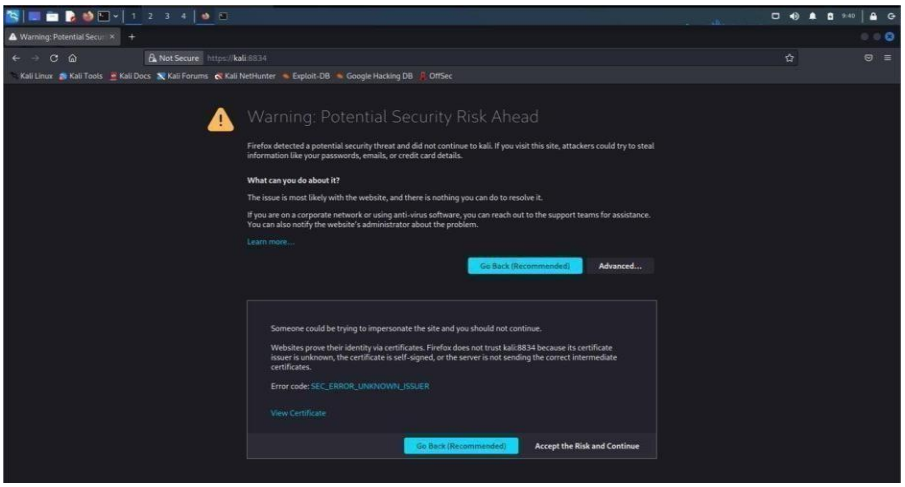
┌──(root㉿kali)-[~]
└─# apt-get install nessusd
Reading package lists ... Done
Building dependency tree ... Done
Reading state information ... Done
E: Unable to locate package nessusd

┌──(root㉿kali)-[~]
└─# curl --request GET \
  --url 'https://www.tenable.com/downloads/api/v2/pages/nessus/files/Nessus-10.7.0-debian10_amd64.deb' \
  --output 'Nessus-10.7.0-debian10_amd64.deb'
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 64.8M    0 64.8M    0     0  6762k      0 --:--:--  0:00:09 --:--:-- 14.1M

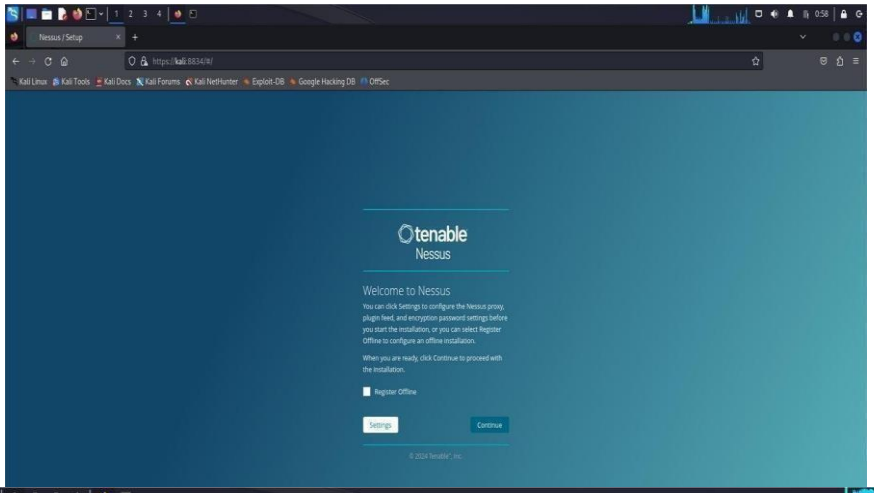
┌──(root㉿kali)-[~]
└─# ls
Desktop  Documents  Downloads  Music  Nessus-10.7.0-debian10_amd64.deb  Pictures  practice  Public  Templates  Videos
```

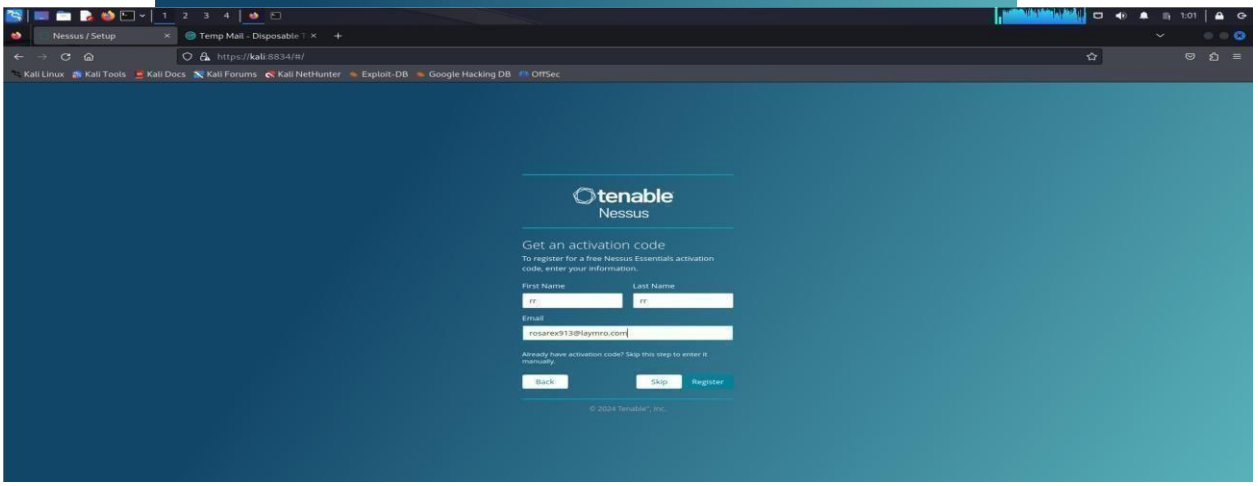
```
┌──(root㉿kali)-[~]
└─# dpkg -i  Nessus-10.7.0-debian10_amd64.deb
Selecting previously unselected package nessus.
(Reading database ... 399450 files and directories currently installed.)
Preparing to unpack Nessus-10.7.0-debian10_amd64.deb ...
Unpacking nessus (10.7.0) ...
Setting up nessus (10.7.0) ...
HMAC : (Module_Integrity) : Pass
SHA1 : (KAT_Digest) : Pass
SHA2 : (KAT_Digest) : Pass
SHA3 : (KAT_Digest) : Pass
TDES : (KAT_Cipher) : Pass
AES_GCM : (KAT_Cipher) : Pass
AES_ECB_Decrypt : (KAT_Cipher) : Pass
RSA : (KAT_Signature) : RNG : (Continuous_RNG_Test) : Pass
Pass
ECDSA : (PCT_Signature) : Pass
ECDSA : (PCT_Signature) : Pass
DSA : (PCT_Signature) : Pass
TLS13_KDF_EXTRACT : (KAT_KDF) : Pass
TLS13_KDF_EXPAND : (KAT_KDF) : Pass
TLS12_PRF : (KAT_KDF) : Pass
PBKDF2 : (KAT_KDF) : Pass
SSHKDF : (KAT_KDF) : Pass
KBKDF : (KAT_KDF) : Pass
```

4. Start the Nessus service with this command:
5. On your browser, go to https://kali:8834/. It would show a warning page.



6. Click on Advanced. Then, click on Accept Risk and Continue.
7. Choose the Nessus Product you prefer. If you want the free version of Nessus, click on Register Nessus Essentials.

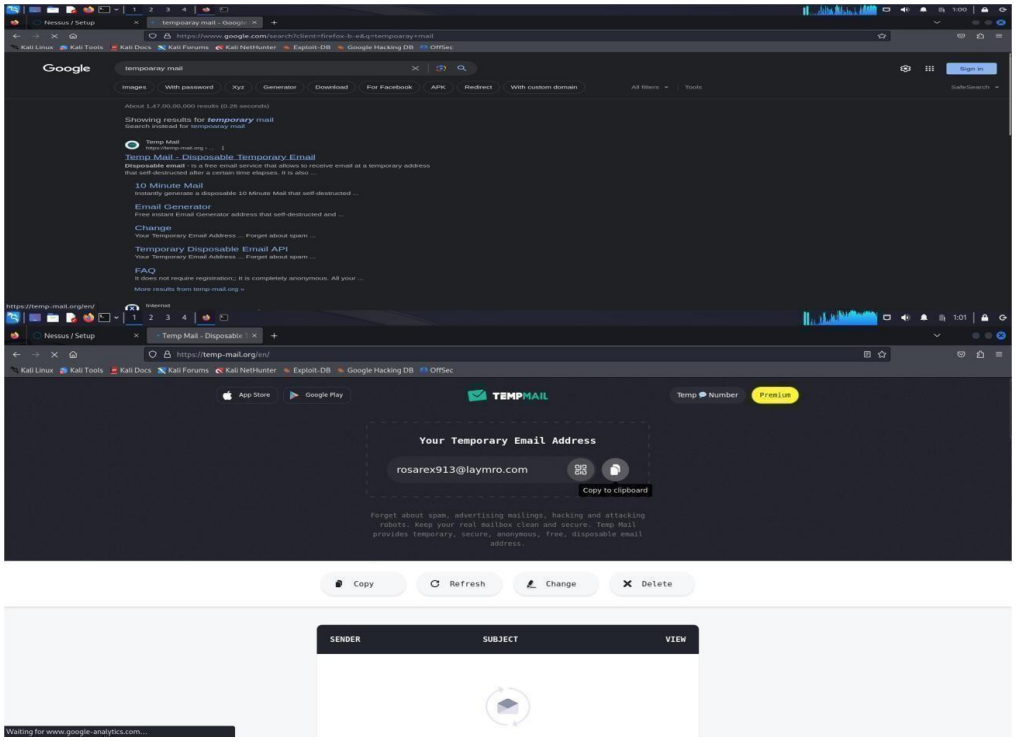


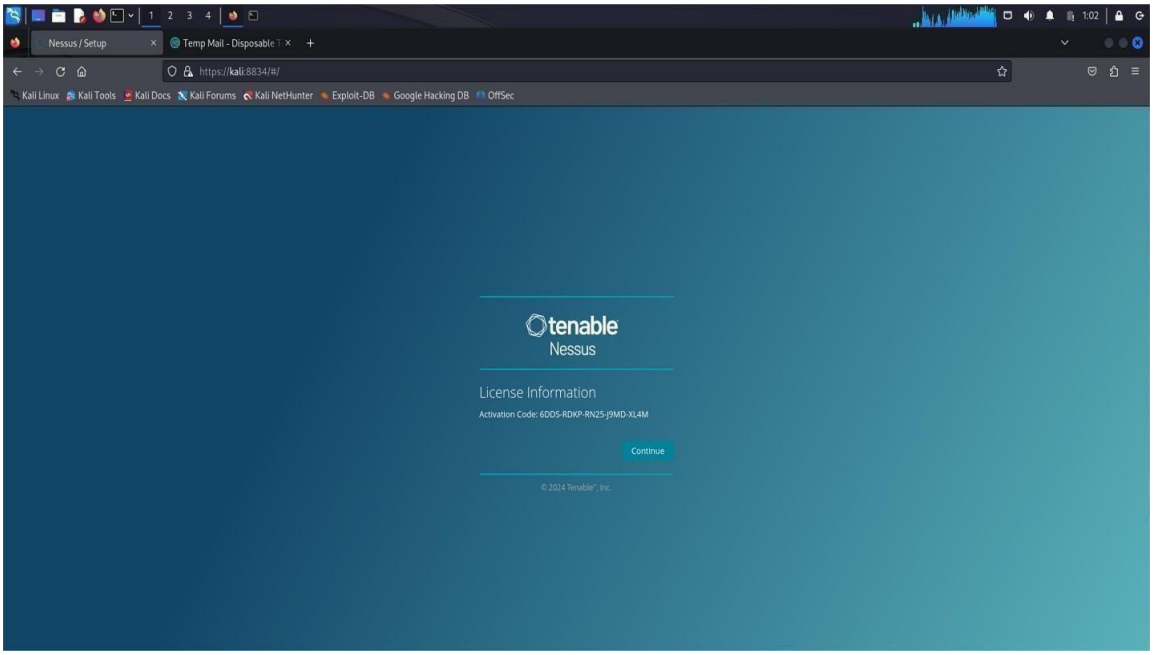


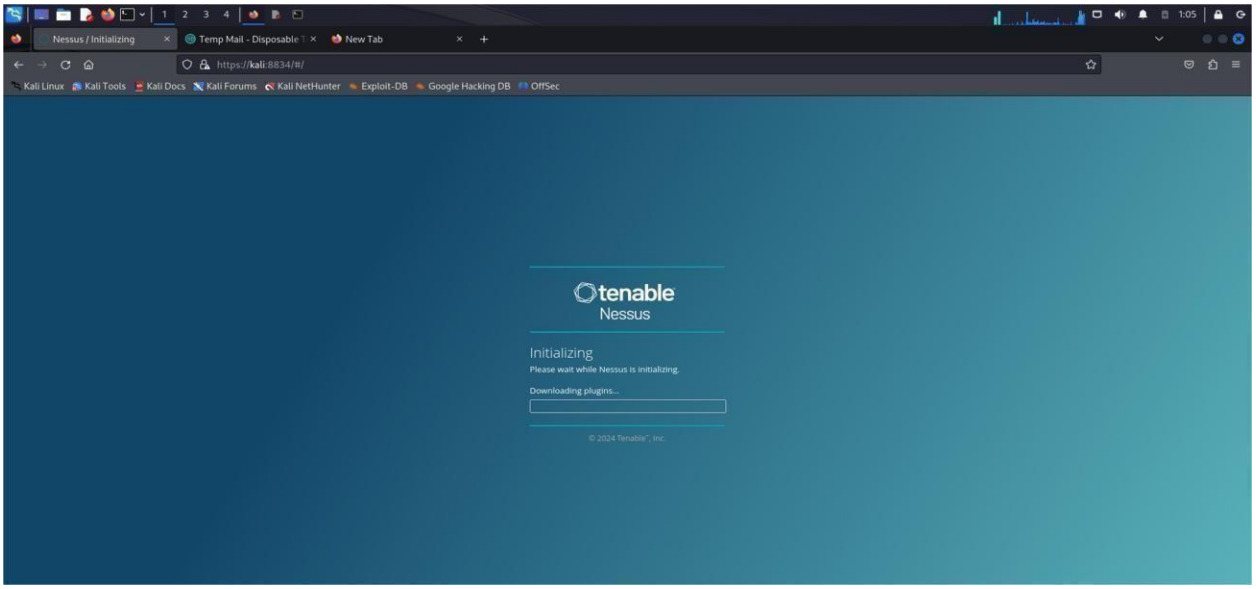
8. Enter your name and email address to receive an activation code by email. Paste the activation code into the space provided and choose a username and password.

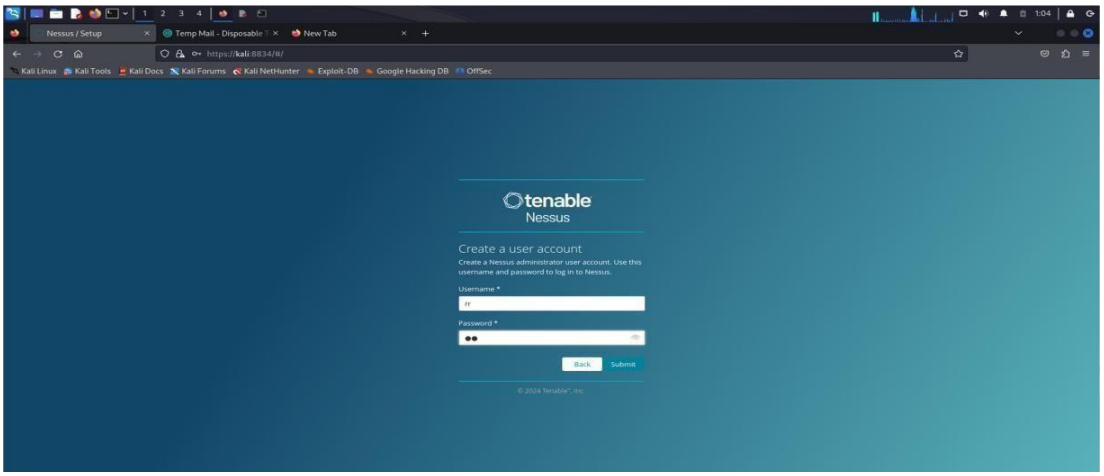
You can use email address as temporary you can visit online temporary email address examples.

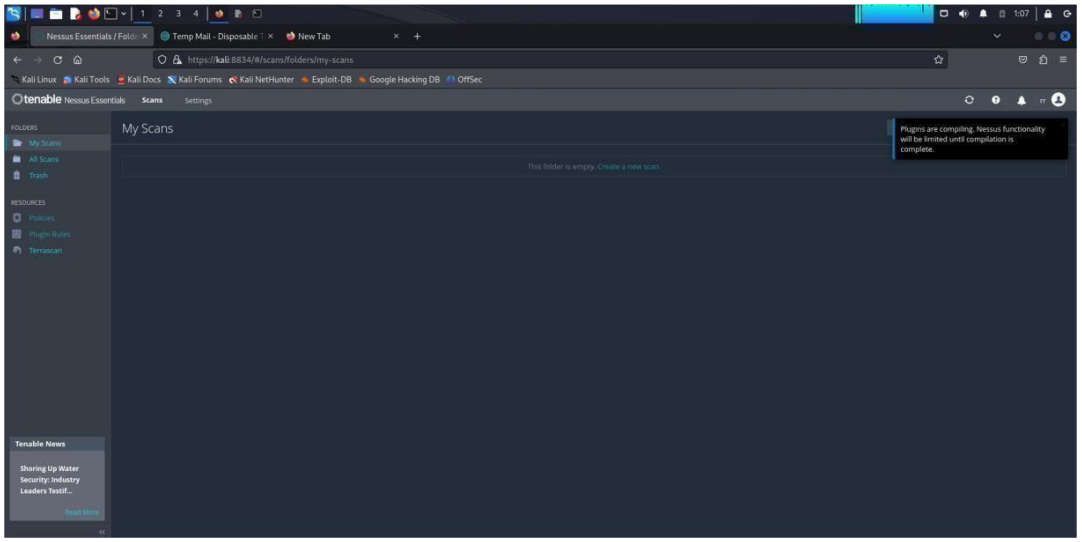
9. Allow Nessus to download the necessary plugins.

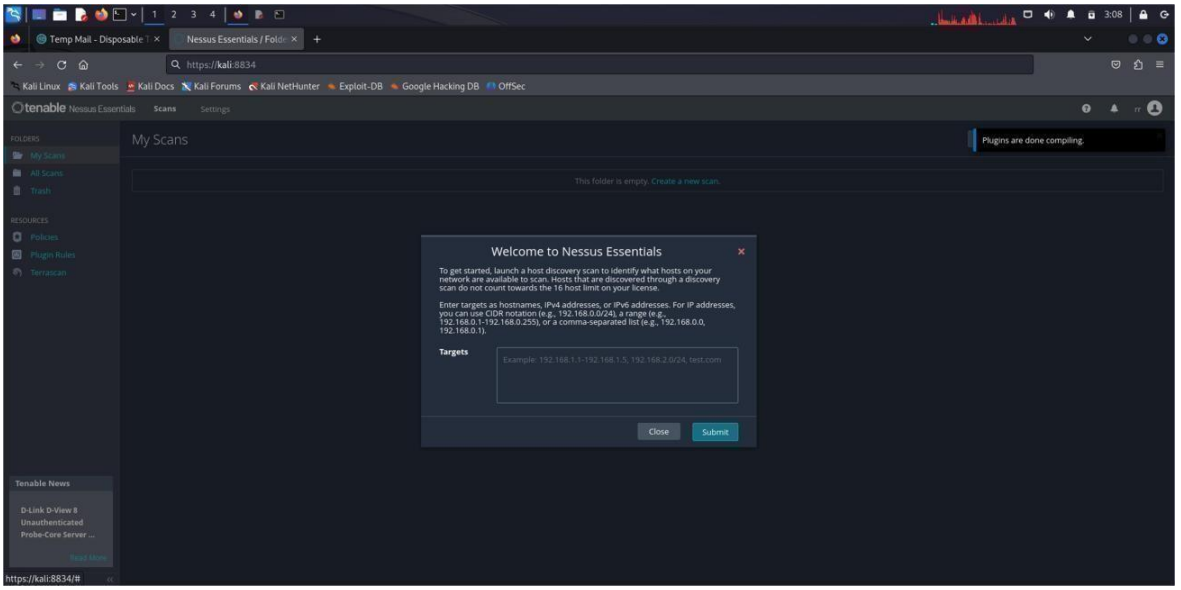



Nessus / Setup
Temp Mail - Disposable
https://kali:8834/#/
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec
tenable
Nessus
License Information
Activation Code: 6DD5-RDKP-RN25-J9MD-XL4M
Continue
© 2024 Tenable™, Inc.

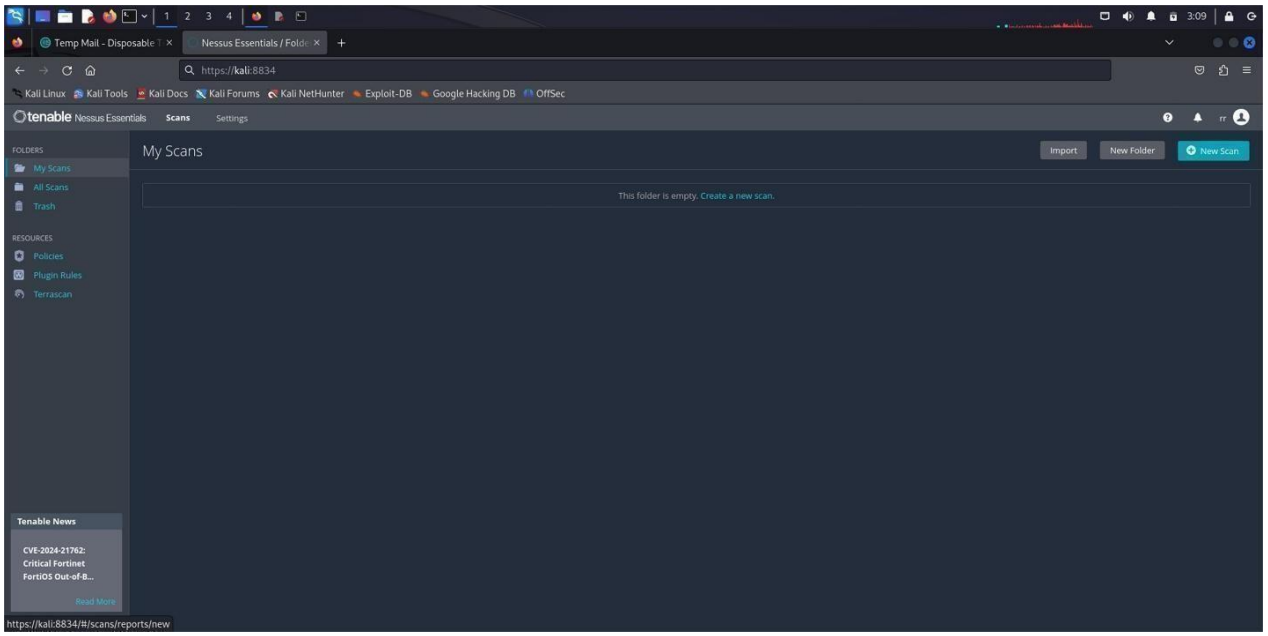

Nessus / Initializing
Temp Mail - Disposable
New Tab
https://kali:8834/#/
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec
tenable
Nessus
Initializing
Please wait while Nessus is initializing.
Downloading plugins...
© 2024 Tenable™, Inc.


Nessus / Setup
Temp Mail - Disposable
New Tab
https://kali:8834/#/
Kali Linux Kali Tools Kali Docs Kali Forums Kali NetHunter Exploit-DB Google Hacking DB OffSec
tenable
Nessus
Create a user account
Create a Nessus administrator user account. Use this username and password to log in to Nessus.
Username *
Password *
Back
Submit
© 2024 Tenable™, Inc.

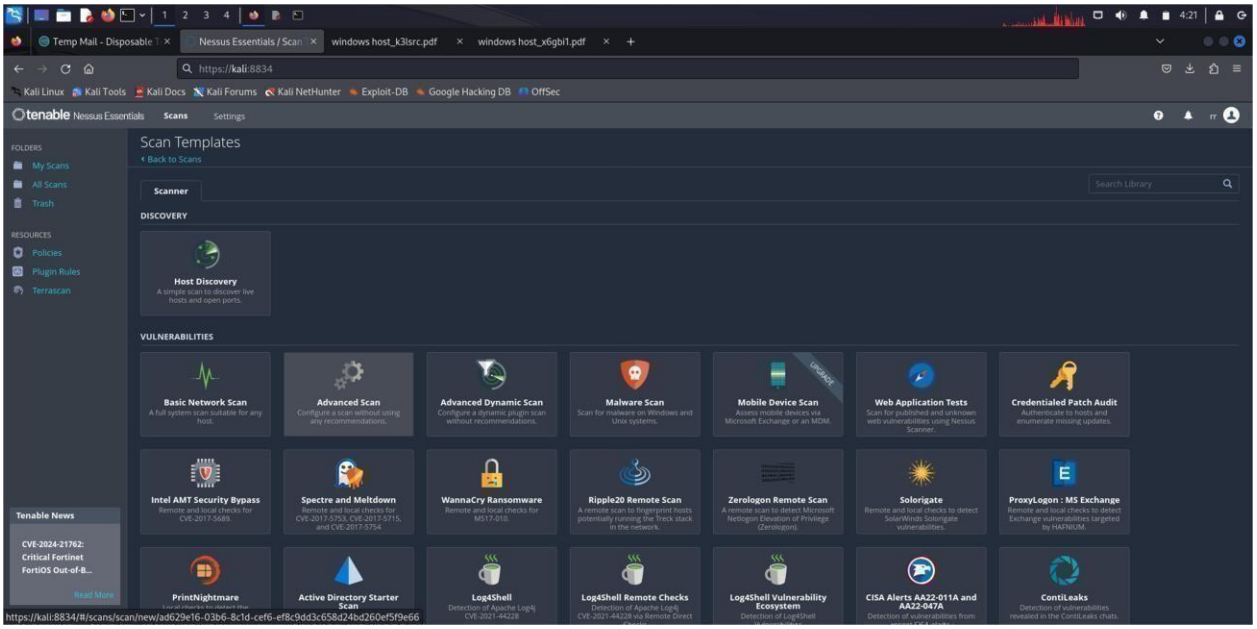

Nessus Essentials / Folde
Temp Mail - Disposable T
New Tab
https://kali:8834/#/scans/folders/my-scans
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
tenable Nessus Essentials
Scans
Settings
FOLDERS
My Scans
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Terrascan
My Scans
This folder is empty. Create a new scan.
Plugins are compiling. Nessus functionality will be limited until compilation is complete.
Tenable News
Shoring Up Water Security: Industry Leaders Testif...
Read More


Temp Mail - Disposable T
Nessus Essentials / Folde
https://kali:8834
Kali Linux
Kali Tools
Kali Docs
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB
OffSec
tenable Nessus Essentials
Scans
Settings
FOLDERS
My Scans
All Scans
Trash
RESOURCES
Policies
Plugin Rules
Terrascan
My Scans
This folder is empty. Create a new scan.
Plugins are done compiling.
Welcome to Nessus Essentials
To get started, launch a host discovery scan to identify what hosts on your network are available to scan. Hosts that are discovered through a discovery scan do not count towards the 16 host limit on your license.
Enter targets as hostnames, IPv4 addresses, or IPv6 addresses. For IP addresses, you can use CIDR notation (e.g., 192.168.0.0/24), a range (e.g., 192.168.0.1-192.168.0.255), or a comma-separated list (e.g., 192.168.0.0, 192.168.0.1).
Targets
Example: 192.168.1.1-192.168.1.5, 192.168.2.0/24, test.com
Close
Submit
Tenable News
D-Link D-View 8 Unauthenticated Probe-Core Server ...
Read More
https://kali:8834/#

Click on **New Scan** to begin scanning for vulnerabilities

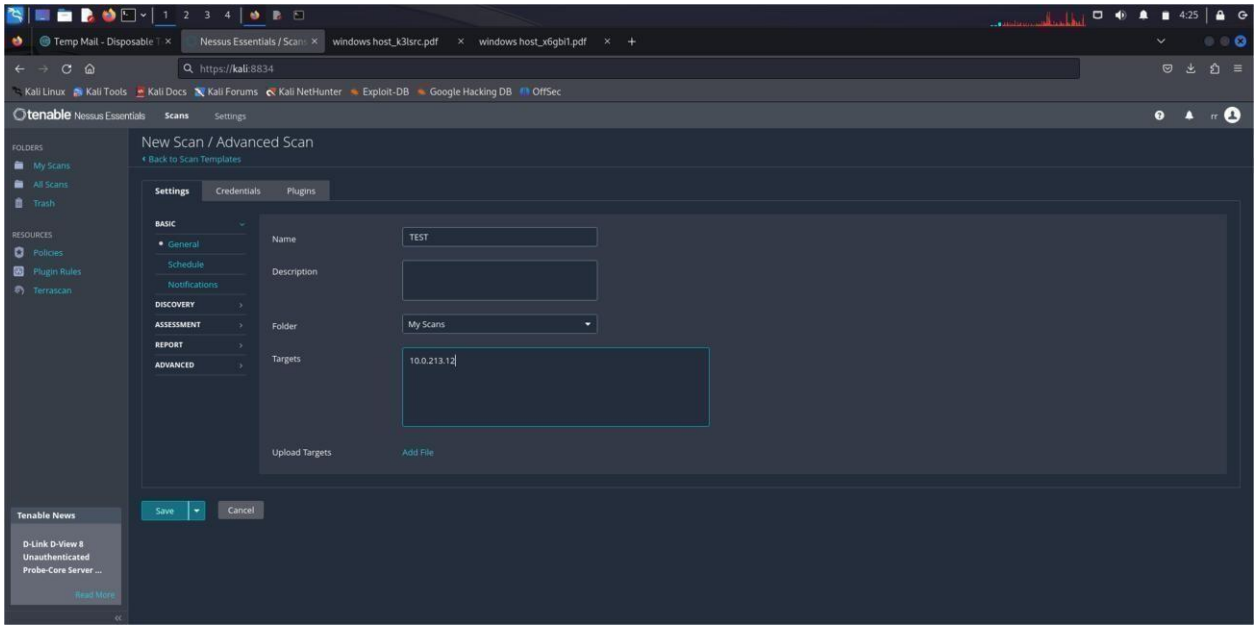


Click on Advance scan



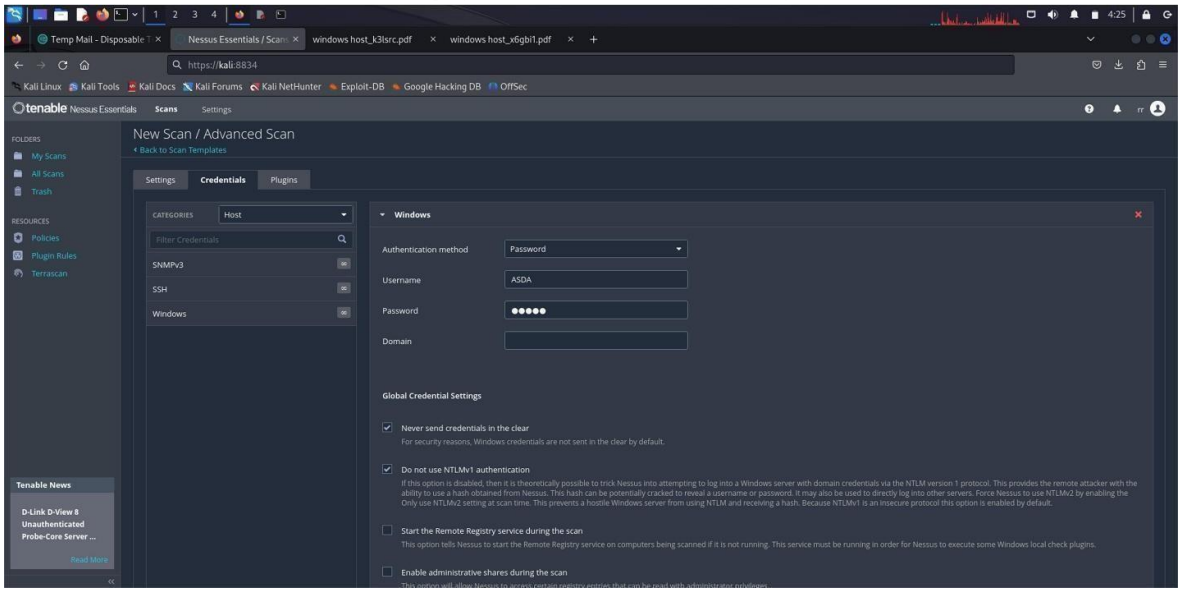
Give name for scaning (e.g test,windows scan..etc)

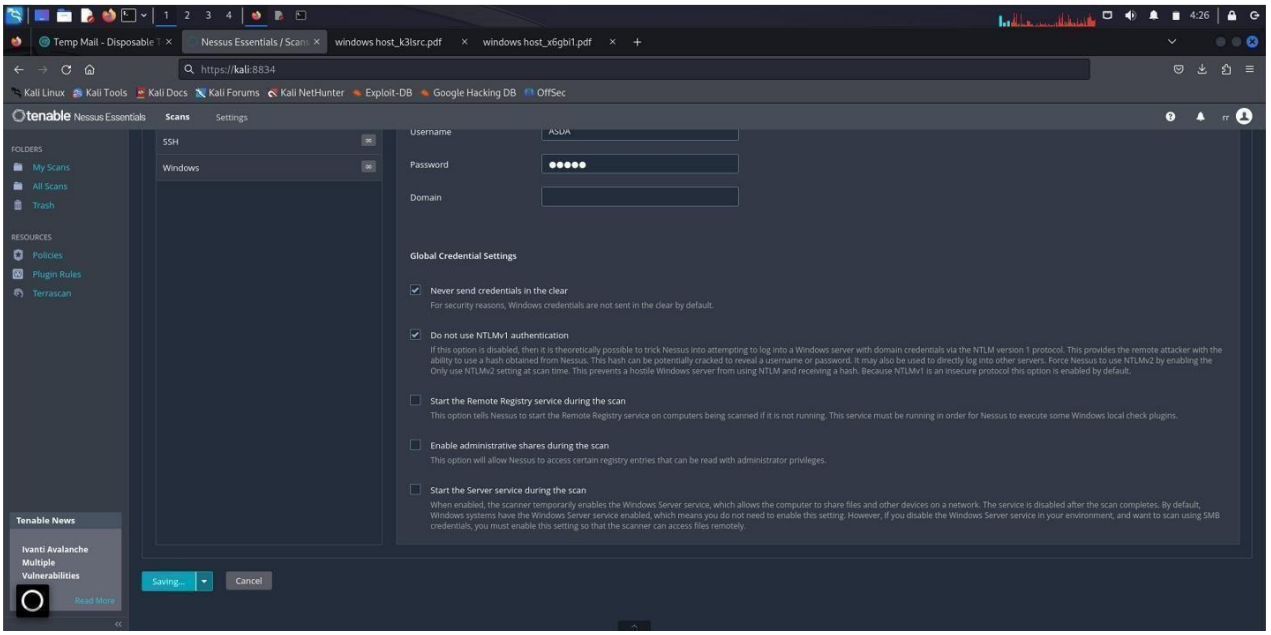
Give the ip address of your windows that you want to scan for vulnerabilities in Target box shown in the below



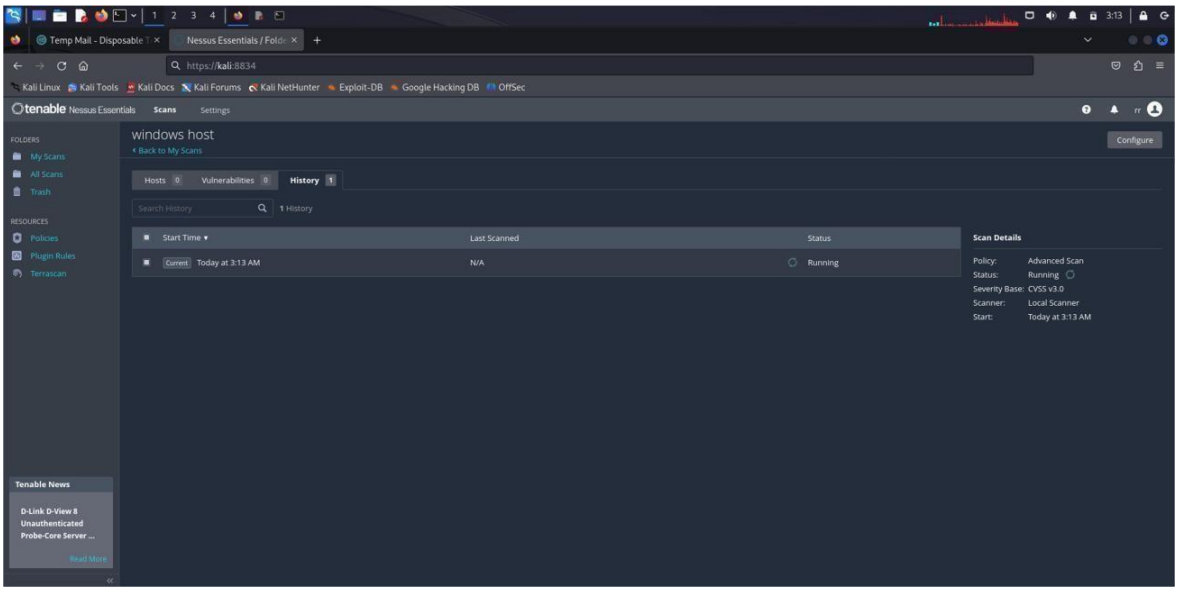
Give the credentials of your windows like username and password if you known (OPTIONAL).

Use of giving credential it will help you scan more into your system.

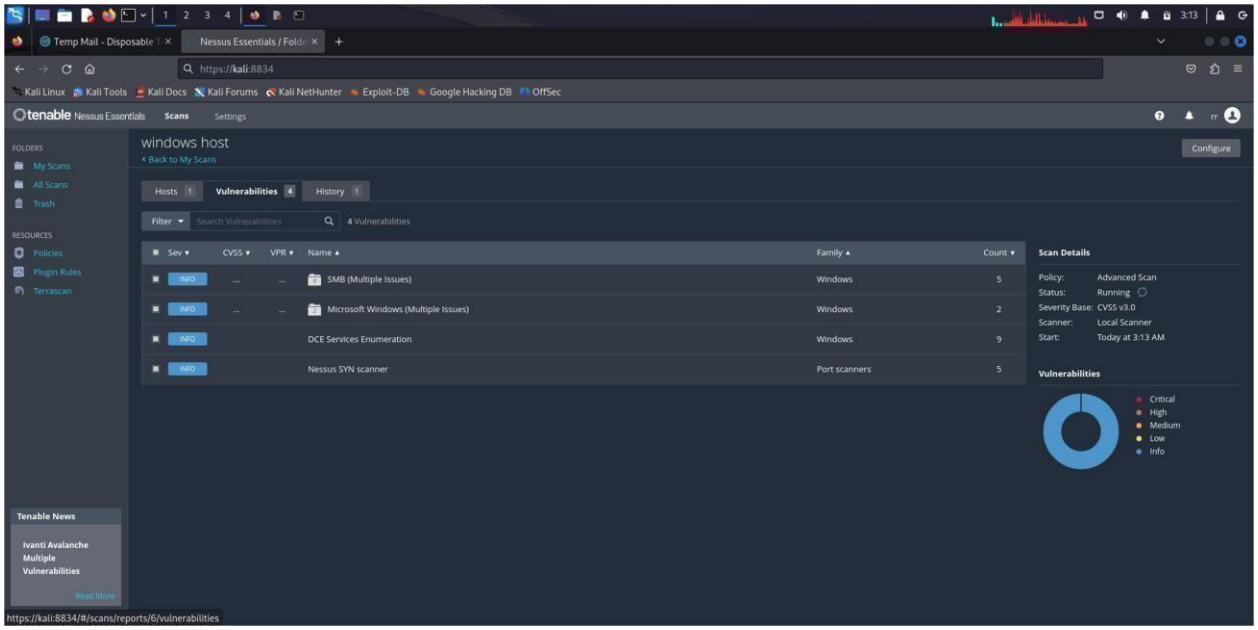


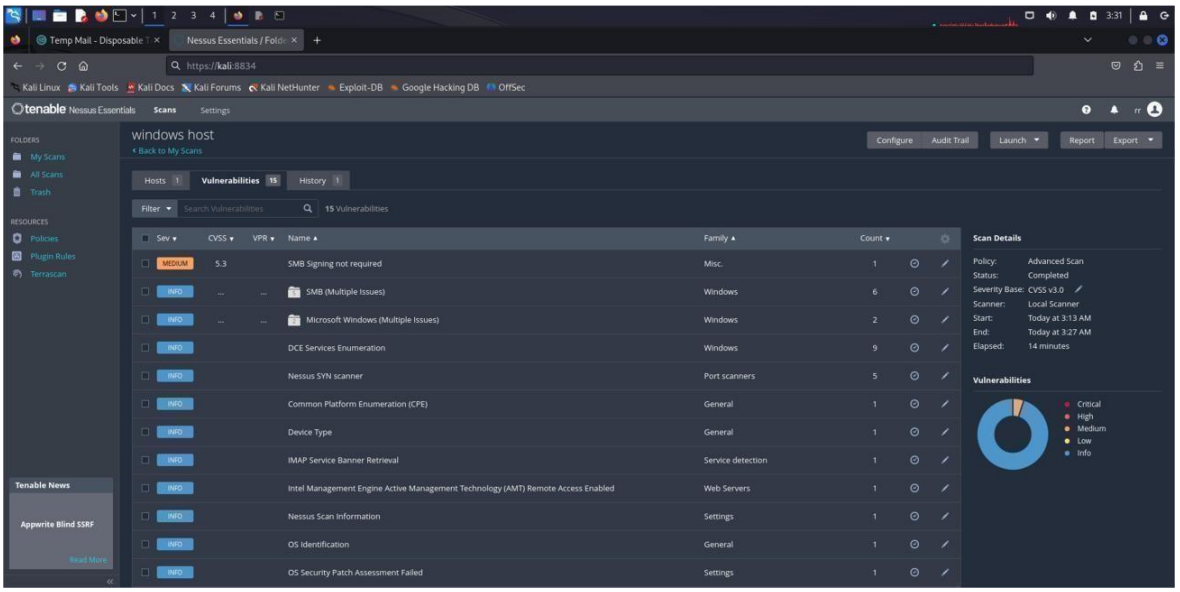


Click on launch:

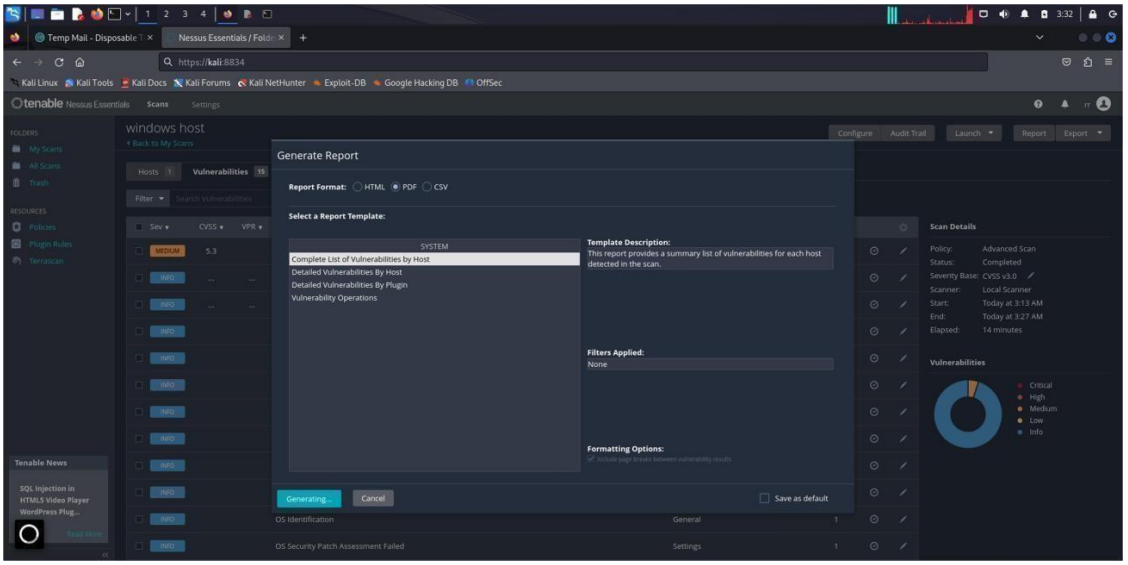


Wait for Some time to get the Output:





ON RIGHT SIDE CLICK ON REPORT TO GENERATE THE ENTIRE REPORT OF YOUR SYSTEM VULNERABILITY:



RESULT:

Thus the IT audit and vulnerability assessment were successfully performed using Nessus. The scan identified system vulnerabilities, misconfigurations, and potential threats. A comprehensive report was generated, helping in understanding the security posture of the system and suggesting mitigation measures were verified successfully.

Ex. No. 5	Implementation of Cyber Forensics tools for Disk Imaging, Data acquisition, Data extraction and Data Analysis and recovery.

AIM:

To Implement Cyber Forensics tools for Disk Imaging, Data acquisition, Data extraction and Data Analysis and recovery.

FTK Imager (Data acquisition and Disk Imaging) SCOPE:

The Scope of practical acquisition of digital evidence can encompass a wide range of methods and devices. Common areas are: Hard Drives, USB, Mobile Devices, Network Forensics, Live system etc.

REQUIREMENTS:

The requirements are Hardware and Devices, FTK imager software, Forensics workstations, sample data, Documentation Templates etc.

THEORY:

Data Acquisition:

- The gathering and recovery of sensitive data during a digital forensic investigation is known as data acquisition and collecting and preserving electronic evidence from various sources.
- Digital forensic analysts need to know how to access, recover, and restore that data as well as how to protect it for future management. This involves producing a forensic image from digital devices and other computer technologies.
- The data acquisition process involves several steps: Identification, Preservation, collection, Verification, Documentation.

Disk Imaging:

Disk imaging, also known as forensic imaging or disk cloning, is a crucial process in digital forensics for cybersecurity. It involves creating a bit-for-bit copy or snapshot of an entire storage device, such as a hard drive, solid-state drive (SSD), or any other media, to preserve its contents for analysis and investigation. The purpose of disk imaging is to obtain an exact replica of the original storage device, including not only the visible files and folders but also the hidden and deleted data.

Types of Disk Imaging:

- **Logical Imaging:** Logical imaging, also known as file-level imaging or logical acquisition, involves creating an image or snapshot of the logical structure of a storage device, such as a file system or a partition. This type of imaging captures the files and directories stored on the disk, along with their metadata and attributes, without including unallocated space or low-level disk structures.

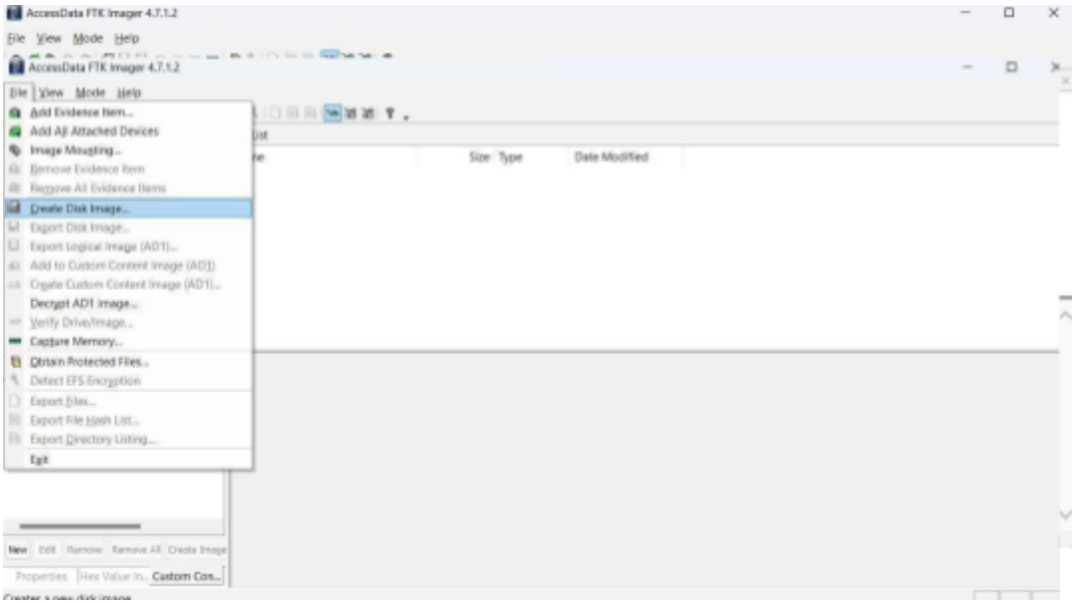
• **Physical Imaging:** Physical imaging, also known as bit-for-bit imaging or sector-level imaging, is a comprehensive process that involves creating an exact copy of the entire storage device, including all sectors, regardless of whether they are allocated or unallocated. This type of imaging captures not only the visible files and folders but also the hidden data, deleted files, slack space, and all other data present on the disk.

Forensic Significance of Imaging:

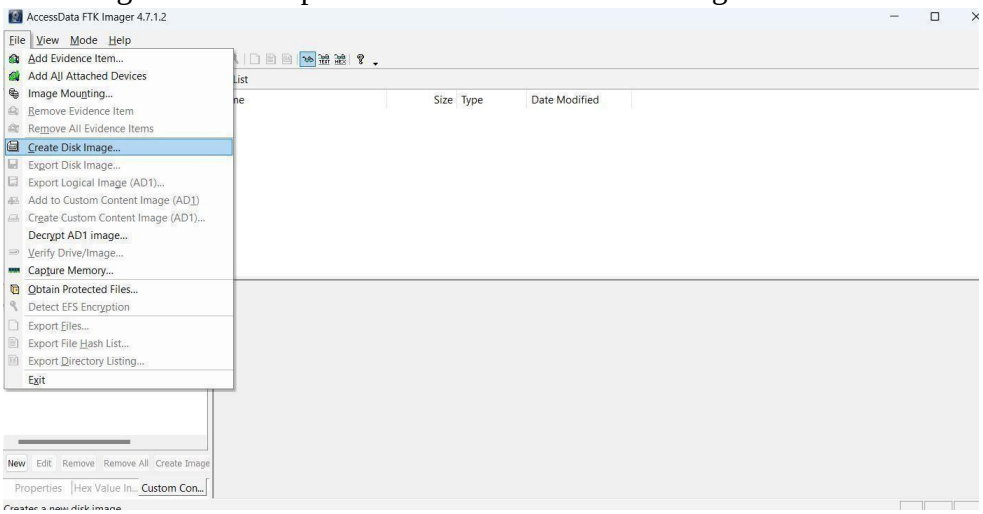
• Forensic imaging prevents the loss of original datas. These imaging tools and techniques are the only way to ensure that electronic data can be successfully admitted as evidence in a court or legal proceeding.

EXAMINATION:

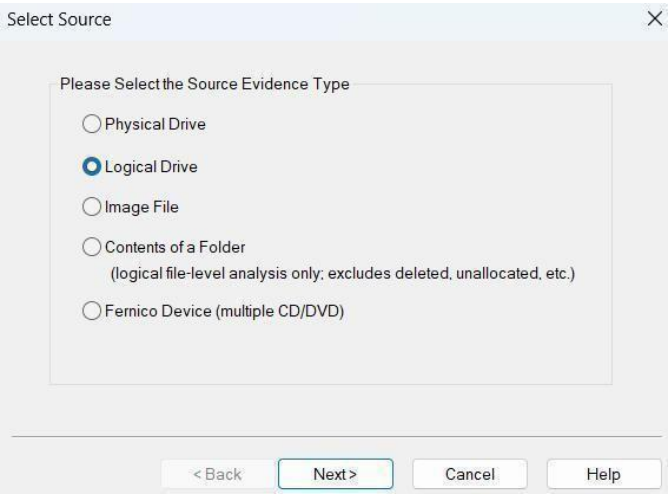
Step 1: Open FTK imager software.



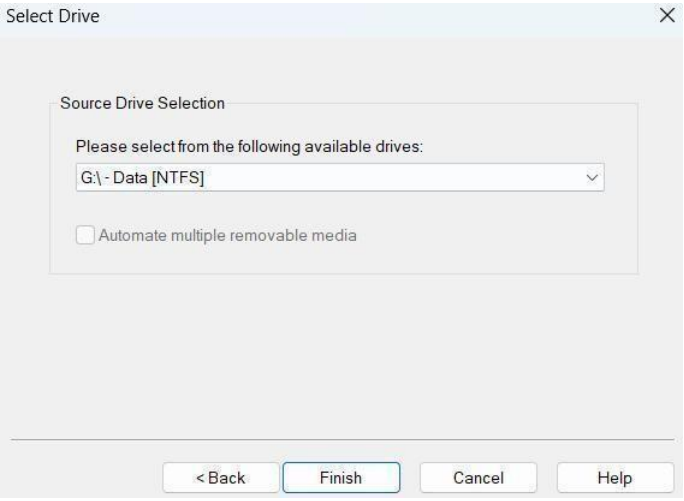
Step 2: Now go to the file option -> choose Create Disk Image.



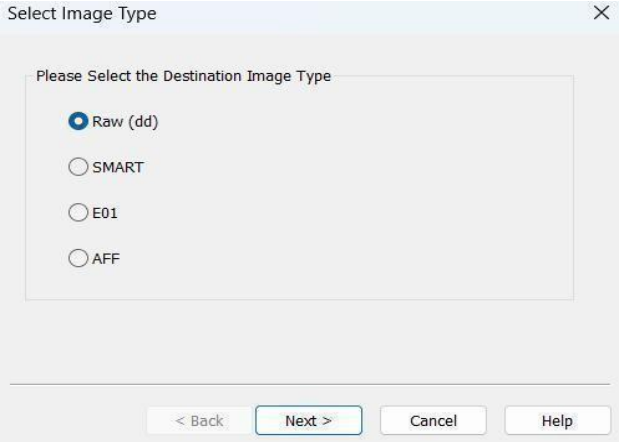
Step 3: After clicking Create Disk image -> select the source of evidence type like Logical Drive, then click next option.



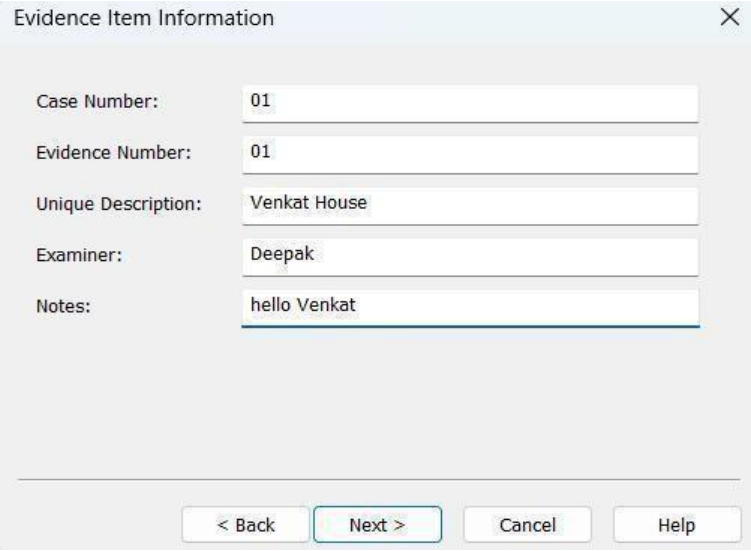
Step 4: Now select source drive path and then press the finish button.



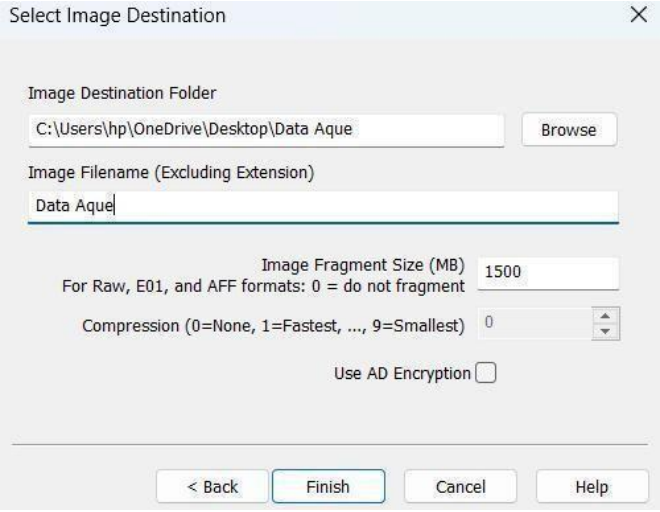
Step 5: Choose destination image type (Raw(dd)) -> Next.



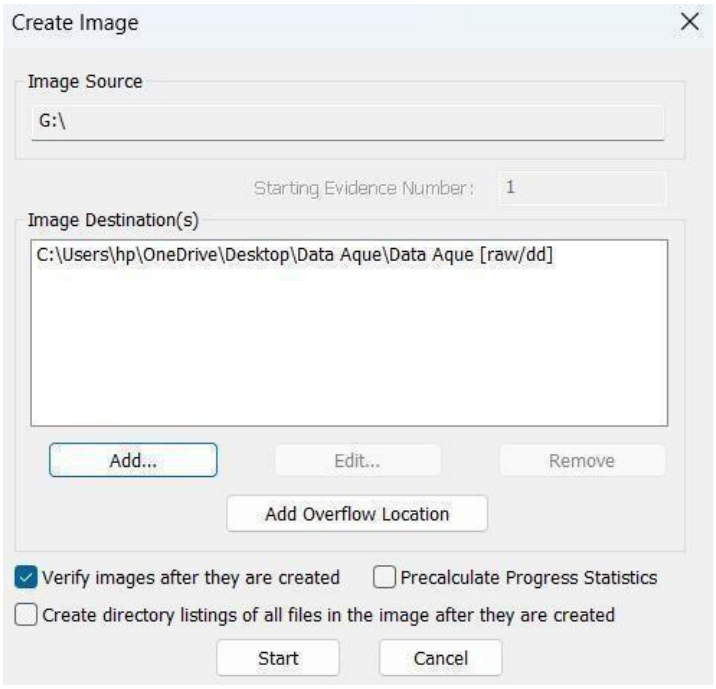
Step 6: Now fill the form of Evidence item information.



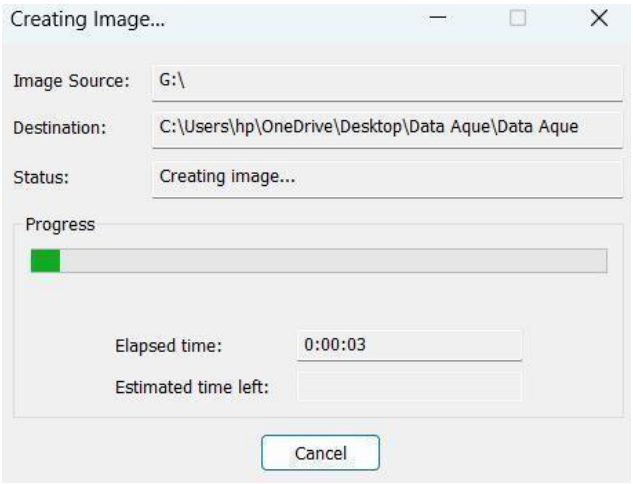
Step 7: After filling the information -> select image destination, enter Image filename and Image fragment size then click the finish button.



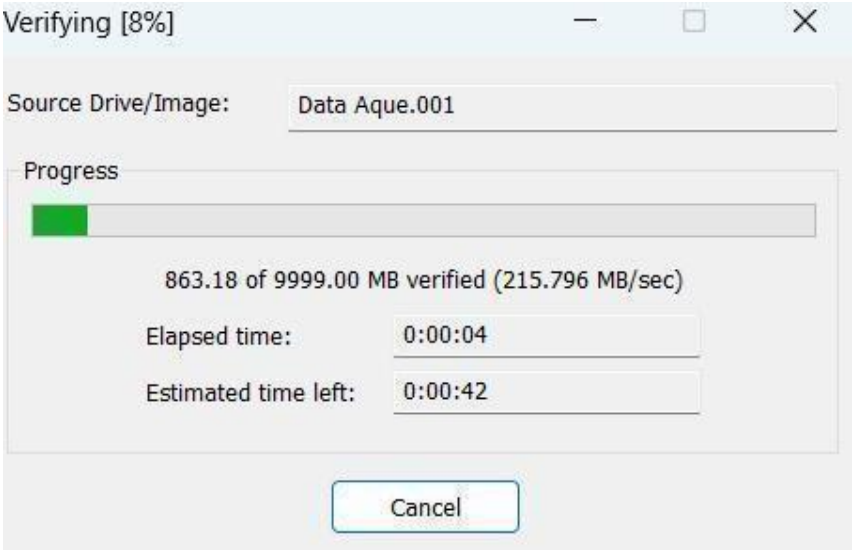
Step 8: After all the processes now, click in the checkbox of verify images after they are created -> start.



Step 9: Progress for creating image.



Step 10: Verifying the source drive/image.



Step 11: Verify result of source drive/image form of hash values.



```
Created By AccessData® FTK® Imager 4.7.1.2

Case Information:
Acquired using: ADI4.7.1.2
Case Number: 01
Evidence Number: 01
Unique description: Venkat House
Examiner: Deepak
Notes: hello Venkat

---------------------------------------------------------------

Information for C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque:

Physical Evidentiary Item (Source) Information:
[Device Info]
 Source Type: Logical
[Drive Geometry]
 Bytes per Sector: 512
 Sector Count: 20,477,952
[Physical Drive Information]
 Removable drive: False
 Source data size: 9999 MB
 Sector count:    20477952
[Computed Hashes]
 MD5 checksum:    a8b13d1099105de415d7b795374ef1aa
 SHA1 checksum:   b473b690eff93531639142ee301187a8a0947d81

Image Information:
 Acquisition started:   Wed Jun 28 13:00:17 2023
 Acquisition finished:  Wed Jun 28 13:01:01 2023
 Segment list:
  C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.001
  C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.002
  C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.003
  C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.004
  C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.005
  C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.006
  C:\Users\hp\OneDrive\Desktop\Data Aque\Data Aque.007

Image Verification Results:
 Verification started:  Wed Jun 28 13:01:01 2023
 Verification finished: Wed Jun 28 13:01:40 2023
 MD5 checksum:    a8b13d1099105de415d7b795374ef1aa : verified
 SHA1 checksum:   b473b690eff93531639142ee301187a8a0947d81 : verified
```

RESULT:

Thus the Cyber forensics analysis was successfully carried out using FTK Imager. Disk imaging and data acquisition were performed, and the integrity of the data was verified using hash values. The process ensured accurate preservation and analysis of digital evidence were verified successfully.

Ex. No. 6	Perform mobile analysis in the form of retrieving call logs, SMS log, all contacts list using the forensics tool like SAFT

AIM:

To perform Mobile Analysis in the form of retrieving call logs, SMS log, all contacts list using the forensics tool like SAFT.

EXPLANATION:

SAFT is a powerful and easy-to-use mobile forensics application developed by well-known security researchers. SAFT allows you to extract valuable information from device in just one click. SAFT allows you to perform logical forensics analysis for android devices. SAFT Free Beta Version supports Android devices and it's free for personal / non-profit usage. You can extract the information described below: Call Logs, SMS Logs, Contacts List. After the investigation of case, SAFT will allow you to generate one-click forensics investigation report.

SAFT allows you to extract valuable information from device in just one click!

FEATURES:

You can extract the information's described below:

- Call Logs
- SMS Logs
- Contacts List
- Keyword Detection
- Reporting

PRICING:

SAFT Community Edition supports Android devices and it's free for personal / non-profit usage.

REMOVAL:

If you used automatic installation:

- Open Control Panel
- Open Add/Remove Programs
- Click on User Profile Hive Service and select Remove

If you manually installed:

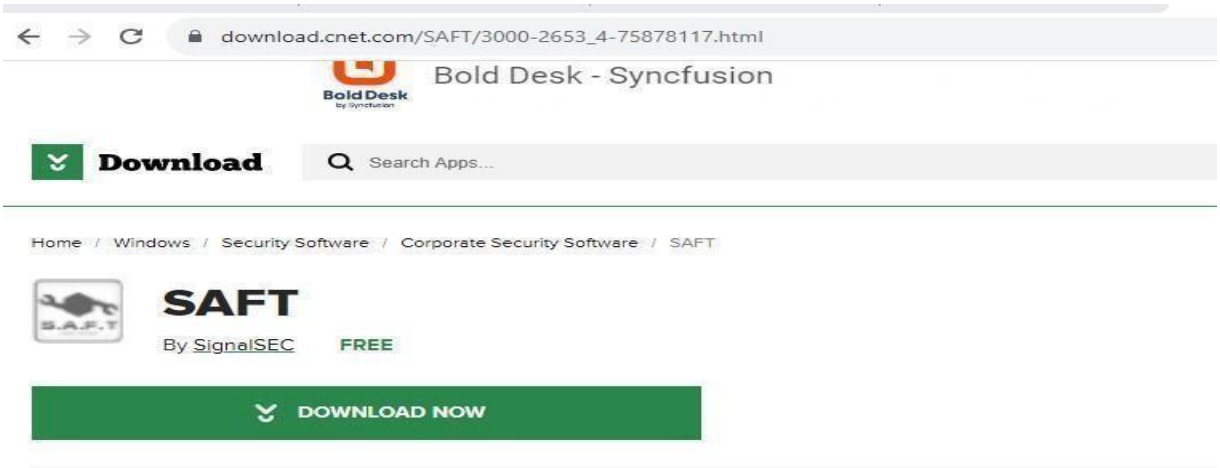
- Run the program with the -remove switch to stop the service and remove it (e.g. uphclean -remove)
- Remove the SAFT directory under c:\program files

SPECIFICATIONS:

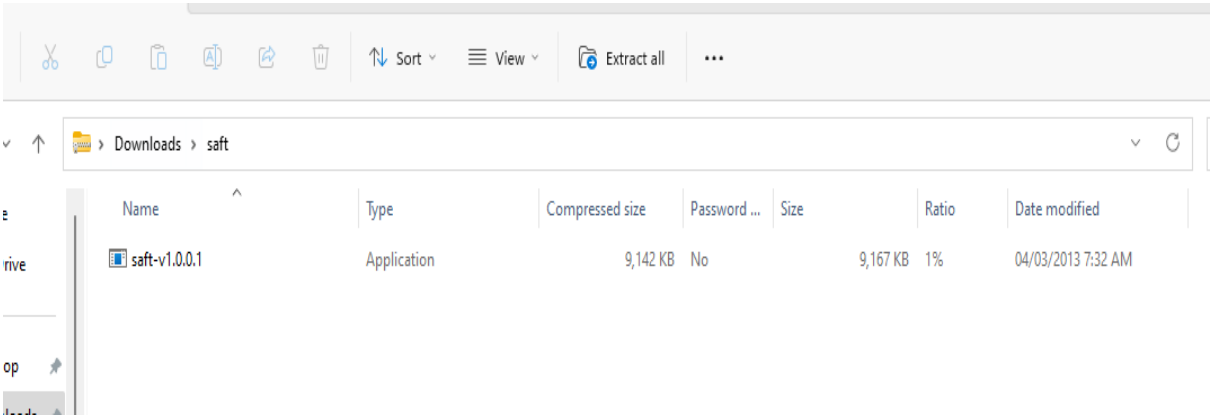
TOOL NAME	OPERATING SYSTEM	LINK
SAFT	Windows 2003, Windows Vista, Windows, Windows 7, Windows XP	https://download.cnet.com/SAFT/3000- 2653_4-75878117.html

SAMPLE VIDEO LINK:

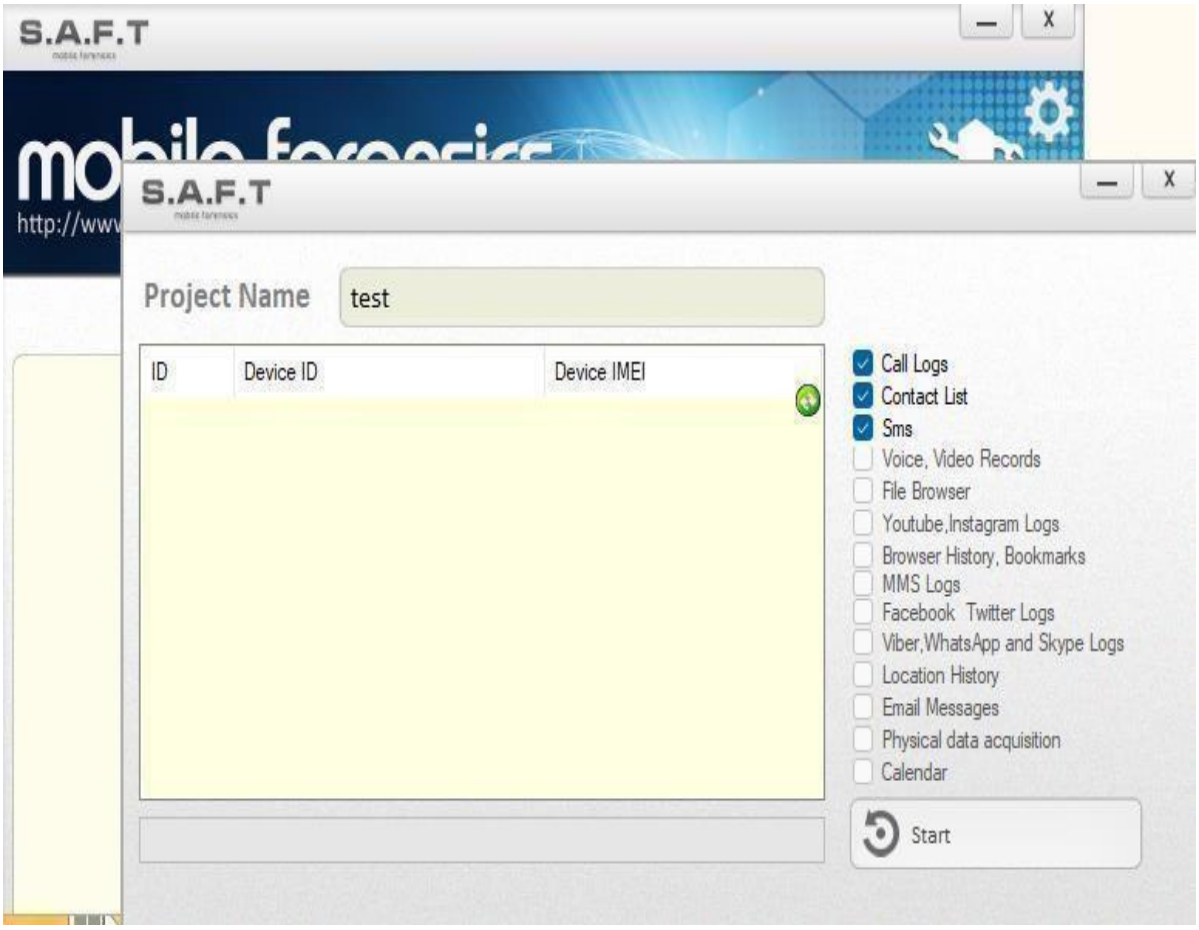
https://www.youtube.com/watch?v=zXiO1vAcKFo



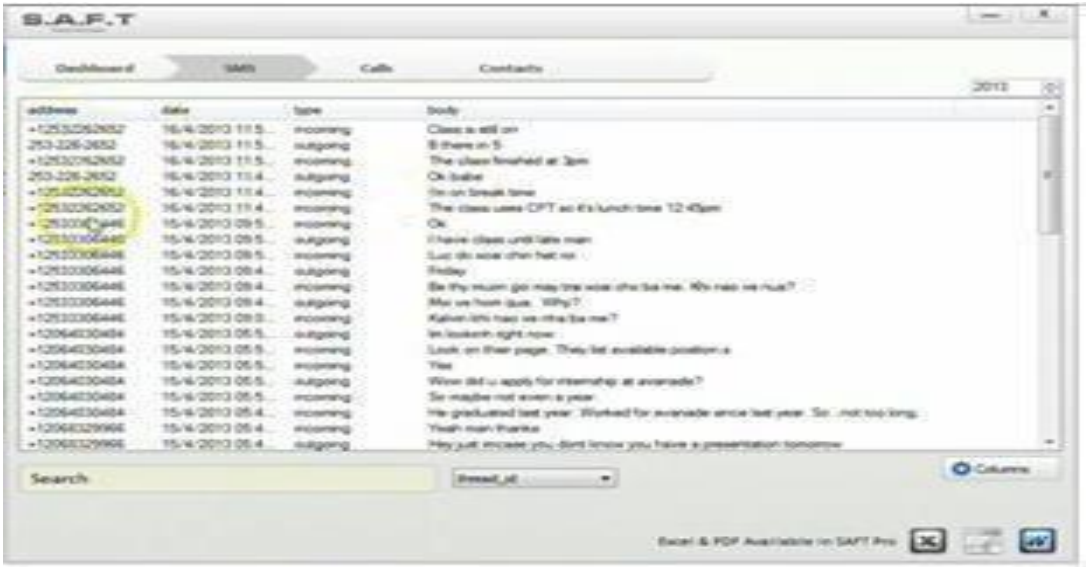
Extract the file and execute the tool and install it



Below is the screenshot of the Dashboard



Below shows about the call logs history





RESULT:

Thus the Mobile forensic analysis was successfully performed using the SAFT tool. Data such as call logs, SMS logs, and contact lists were extracted efficiently. The tool generated a structured report, demonstrating its capability in mobile data investigation was verified successfully.

Ex. No. 7	Implementation to identify web vulnerabilities using OWASP project.

AIM:

To Implement the Identification of web vulnerabilities using OWASP project.

OBJECTIVE:

To analyze a web application for security weaknesses such as misconfigurations, insecure inputs, and common vulnerabilities using an automated security scanner.

REQUIREMENTS:

- Kali Linux / Ubuntu
- Internet connection
- OWASP ZAP (ZAP Proxy tool)
- Target website (test site like DVWA or any demo site)

INTRODUCTION:

- OWASP ZAP (Zed Attack Proxy) is an open-source web application security scanner.
- It helps in identifying vulnerabilities during the development and testing phases.
- It can be used for:
- Automated vulnerability scanning
- Manual testing
- Intercepting HTTP/HTTPS requests
- It is widely used by:
- Security analysts
- Penetration testers
- Developers

IMPLEMENTATION:

Step 1: Update system

sudo apt-get update

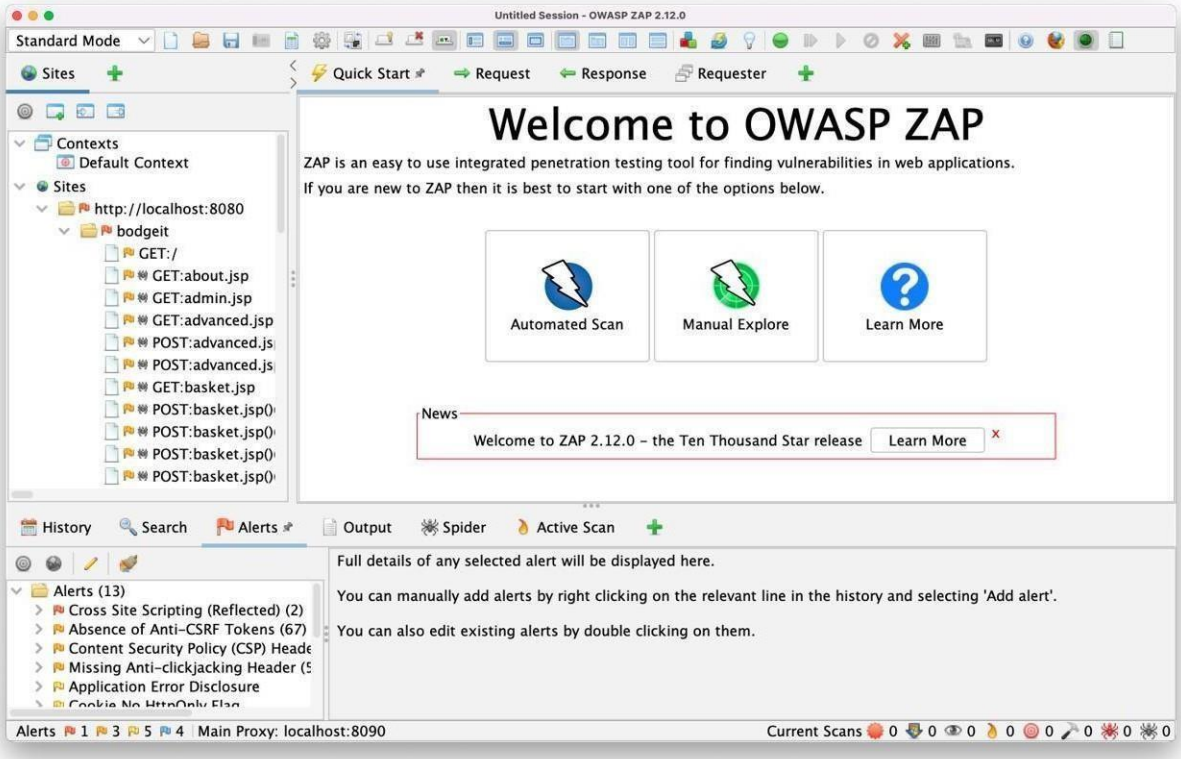
Step 2: Install OWASP ZAP

sudo apt-get install zaproxy

Step 3: Launch OWASP ZAPzaproxy

Step 4: Start a New Session

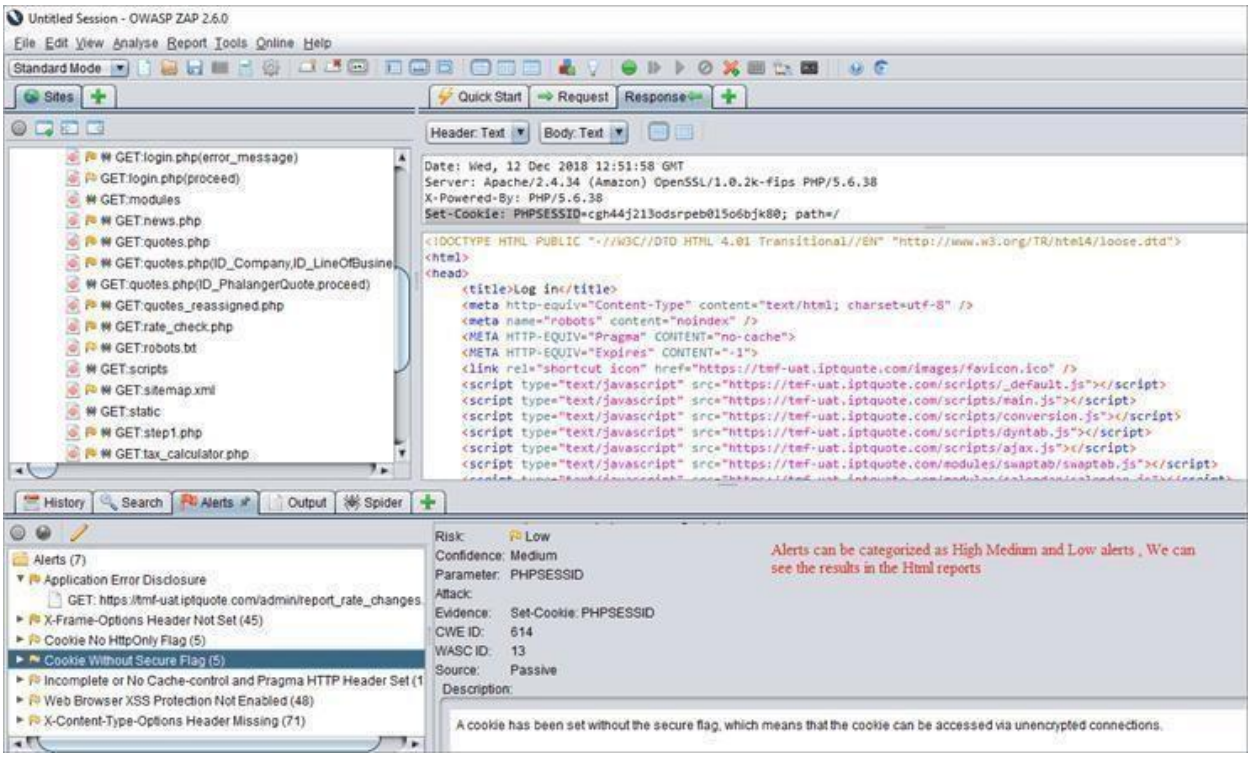
- Open ZAP
- Click on “Start” or create a new session
- Choose “No, I do not want to persist this session” (optional)



Step 5: Enter Target URL

In the URL bar, enter the target website Example:

http://testphp.vulnweb.com



Step 6: Perform Automated Scan

- Click on “Automated Scan”
- Enter target URL
- Click Attack

Step 7: Spidering (Crawling)

ZAP explores the website pages Identifies all endpoints and links

Step 8:

- Active Scanning
- ZAP sends test payloads Detects vulnerabilities like:
  - XSS
  - SQL Injection
- Directory traversal Step 9: Analyze Results
- Alerts are categorized as:
  - High
  - Medium
  - Low
  - Informational

Click on each alert to view:

- Description
- Risk level
- Solution

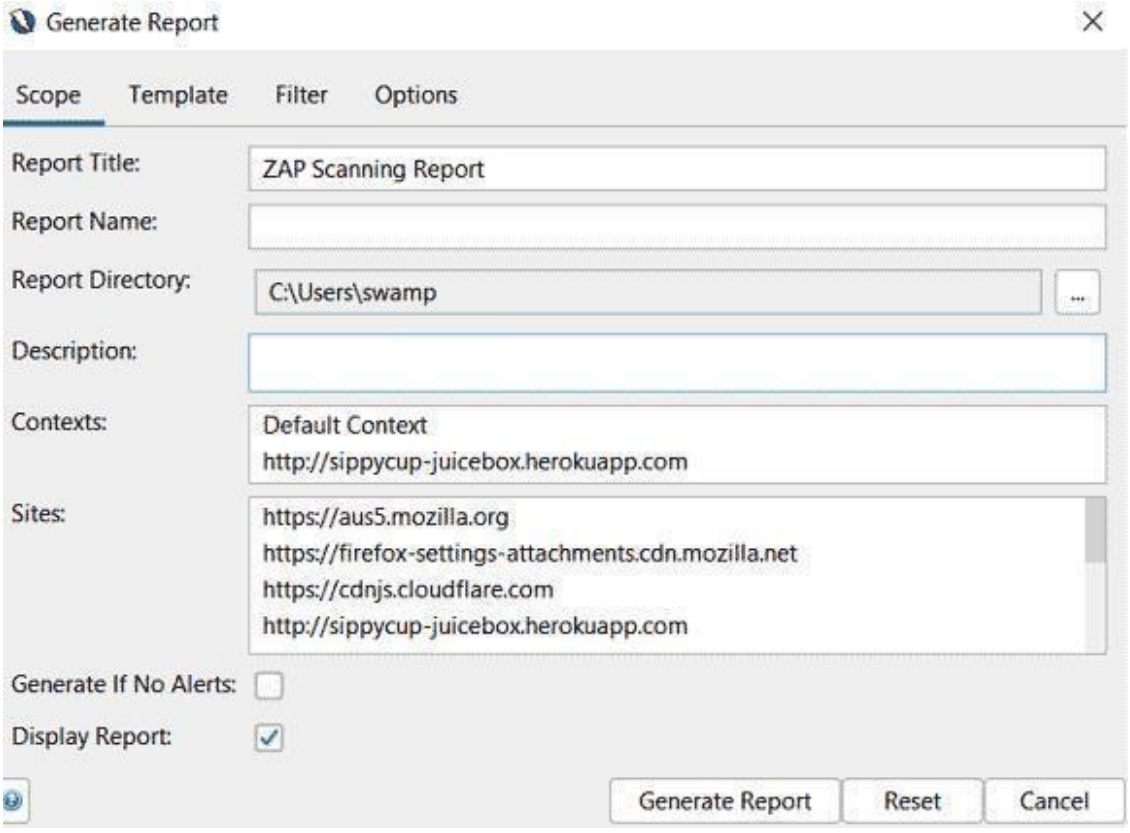
Step 10: Generate Report

Go to Report → Generate Report Save in HTML/PDF format

OUTPUT / OBSERVATION

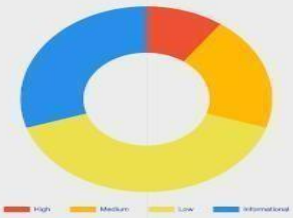
The scan successfully identified vulnerabilities in the target web application. Alerts were categorized based on severity levels.

Detailed reports provided information about risks and recommended fixes.



ZAP Scanning Report

Generated on Fri, 4 Feb 2022 12:07:12



Summary of Alerts

Risk Level	Number of Alerts
High	2
Medium	4
Low	8
Informational	6

Alerts

Name	Risk Level	Number of Instances
Anti-CSRF Tokens Check	High	10
Cross Site Scripting (Reflected)	High	2
Buffer Overflow	Medium	529
Content Security Policy (CSP) Header Not Set	Medium	58
Example Passive Scan Rule: Denial of Service	Medium	7
X-Frame-Options Header Not Set	Medium	55
Absence of Anti-CSRF Tokens	Low	73
Application Error Disclosure	Low	1
Cookie No HttpOnly Flag	Low	1
Cookie without SameSite Attribute	Low	2
In Page Banner Information Leak	Low	3
Information Disclosure - Debug Error Messages	Low	1
Permissions Policy Header Not Set	Low	59
X-Content-Type-Options Header Missing	Low	62
Information Disclosure - Suspicious Comments	Informational	58
Loosely Scoped Cookie	Informational	3
Modern Web Application	Informational	33
Non-Storable Content	Informational	2
Storable and Cacheable Content	Informational	64
User Controllable HTML Element Attribute (Potential XSS)	Informational	32

Passing Rules

Name	Rule Type	Threshold	Strength
Remote Code Execution - Shell Shock	Active	MEDIUM	MEDIUM
Directory Browsing	Active	MEDIUM	MEDIUM
CRLF Injection	Active	MEDIUM	MEDIUM
Path Traversal	Active	MEDIUM	MEDIUM
Remote File Inclusion	Active	MEDIUM	MEDIUM
Parameter Tampering	Active	MEDIUM	MEDIUM
Server Side Include	Active	MEDIUM	MEDIUM
Session Fixation	Active	MEDIUM	MEDIUM
Cross Site Scripting (Persistent)	Active	MEDIUM	MEDIUM
Script Active Scan Rules	Active	MEDIUM	MEDIUM
Cross Site Scripting (Persistent) - Prime	Active	MEDIUM	MEDIUM
Cross Site Scripting (Persistent) - Spider	Active	MEDIUM	MEDIUM
SQL Injection	Active	MEDIUM	MEDIUM
SQL Injection - MySQL	Active	MEDIUM	MEDIUM
SQL Injection - Hypersonic SQL	Active	MEDIUM	MEDIUM
ELMAH Information Leak	Active	MEDIUM	MEDIUM
.htaccess Information Leak	Active	MEDIUM	MEDIUM
Server Side Code Injection	Active	MEDIUM	MEDIUM
Remote OS Command Injection	Active	MEDIUM	MEDIUM
Source Code Disclosure - Git	Active	MEDIUM	MEDIUM
Source Code Disclosure - File Inclusion	Active	MEDIUM	MEDIUM

RESULT:

Thus the Web vulnerabilities were successfully identified using OWASP ZAP. The tool effectively scanned the application and detected security issues, helping in improving the overall security of the web application was successfully verified.

Ex. No. 8	Configuration and Experimentation of Firewall and Proxy Services using iptables and Squid on Kali Linux

AIM:

To configure and experiment with firewall and proxy packages like iptables and Squid on Kali Linux.

OBJECTIVE:

- To understand the working of firewall rules using iptables
- To configure packet filtering and traffic control
- To install and configure Squid proxy server
- To monitor and restrict network access using proxy rules
- To analyze the effect of firewall and proxy configurations on network traffic

REQUIREMENTS:

- System with Kali Linux installed
- Root or sudo privileges
- Internet connectivity
- Packages:
- iptables
- squid
- Basic knowledge of Linux commands

PROCEDURE:

Step 1: Update the system

Open terminal and run:

sudo apt update && sudo apt upgrade -y



Step 2: Install required packages

Install iptables and squid using:

sudo apt install iptables squid -y



Step 3: Configure iptables firewall

First, check existing rules:

sudo iptables -L

To block incoming traffic from a specific IP address:

sudo iptables -A INPUT -s 192.168.1.10 -j DROP

To allow HTTP traffic (port 80):

sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT

Save the firewall rules:

sudo iptables-save > /etc/iptables/rules.v4

Step 4: Configure Squid proxy

Open squid configuration file:

sudo nano /etc/squid/squid.conf

Add the following lines to allow local network:

acl localnet src 192.168.1.0/24

http_access allow localnet

Set proxy port:

http_port 3128

To block a website:

acl blocked_sites dstdomain .facebook.com

http_access deny blocked_sites

Save and exit the file.

Step 5: Restart Squid service

sudo systemctl restart squid

Step 6: Configure browser proxy

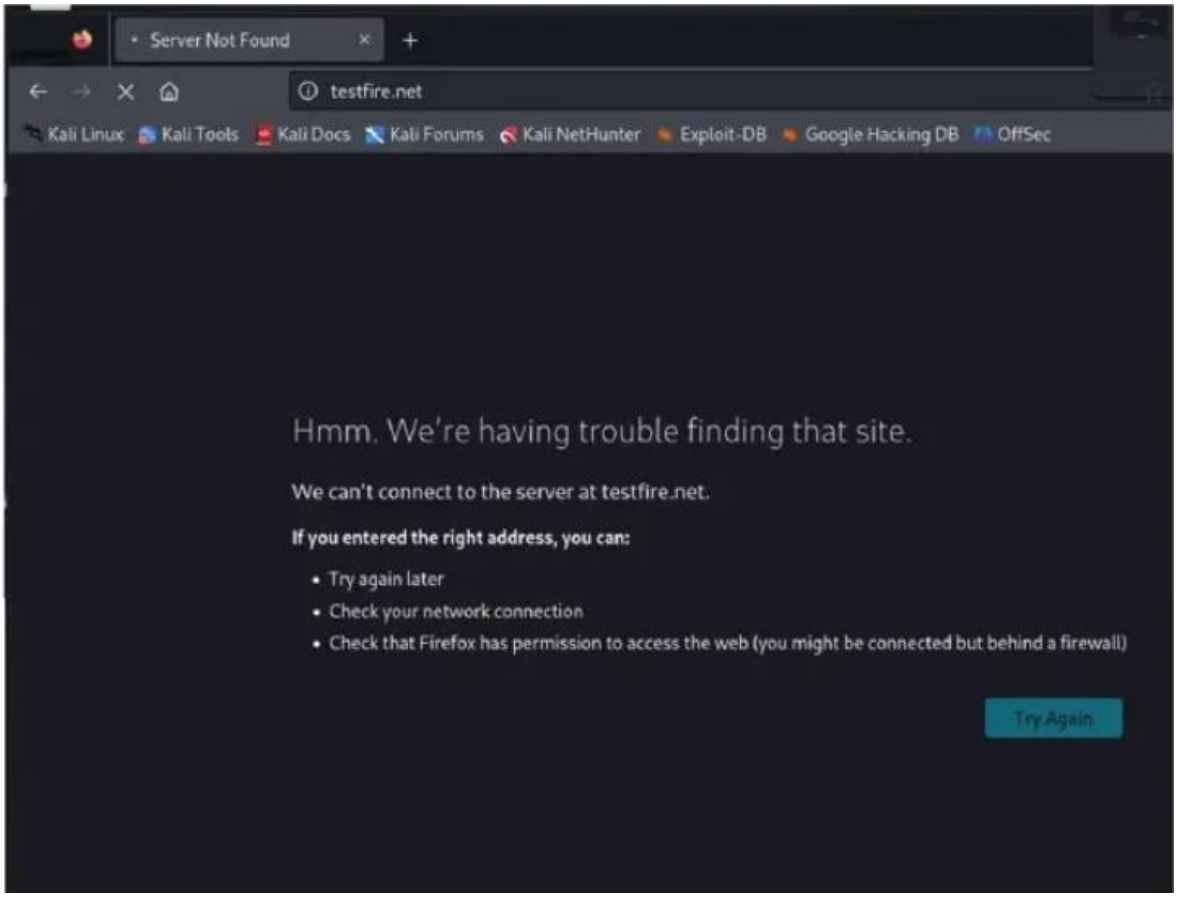
Set proxy IP as 127.0.0.1 and port as 3128 in browser settings

Step 7: Testing

Try accessing allowed and blocked websites

Verify firewall rules using:

sudo iptables -L



RESULT:

Thus, theFirewall rules using iptables and proxy configuration using Squid were successfully implemented on Kali Linux. The system effectively controlled and monitored network traffic by allowing, blocking, and filtering access based on defined rules were verified successfully.